DARKWOODS FOREST CARBON PROJECT

Document Prepared by:

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Project Title	Darkwood Forest Carbon Project
Version	v1.0
Date of Issue	05-August-2019
Project Location	Canada, British Columbia
Project Proponent(s)	Nature Conservancy of Canada Rob Wilson rob.wilson@natureconservancy.ca +1 800 465 0029
Assessor Contact	Aster Global Environmental Solutions Shawn McMahon smcmahon@asterglobal.com +1 330 833 9941
Project Lifetime	01 April 2008 – 01 April 2107; 100-year lifetime.
History of SD VISta Status	New SD VISta project
Other Certification Programs	Verified Carbon Standard Project ID: 607. Climate, Community & Biodiversity Standard Project ID: 607.

Expected Future Assessment Schedule	Initial Validation/Verification: Fall 2019.

CONTENTS

1 SUMMARY OF SDG CONTRIBUTIONS

Table 1. Summary of Project SDG Contributions

Row number	Estimated Project Contribution by the End of Project Lifetime	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Section Reference	Claim, Asset or Label
Sequential row number	*Brief description of the quantifiable impact of the project's activities related to the SDG indicator*	*SDG target number*	*SDG indicator number, where directly related to an indicator, and exact text of indicator*	*Indicate the project's net expected contribution to the indicator*		
1)	Carbon Emissions Reductions - By conserving 54,792 ha of inland temperate rainforest, the project will reduce carbon emissions and sequester carbon by ~14.66 million tCO_2e over 100 years.	13.0	Tonnes of greenhouse gas emissions avoided or removed	Increase	VCS & CCB validation report.	SD VISta - labeled VCU
2)	Forest Biomass – by managing 54,792 ha of forest for conservation purposes, the project will retain and increase the aboveground biomass stocks on the project area.	15.2	15.2.1	Increase	VCS validation report	Claim

3)	Water -	6.1		Increase		Claim
4)	Biodiversity	15.5		Implemented activities to increase.	CCB Validation Report	Claim
5)						

2 PROJECT DESIGN

2.1 Project Objectives, Context and Long-term Viability

2.1.1 Summary of Project Sustainable Development Objective(s)

The Nature Conservancy of Canada (NCC) acquired the fee simple 54,792 ha (135,394 acre) Darkwoods property near Creston, British Columbia, Canada from the Pluto Darkwoods Corporation in April of 2008, with the objective of managing the land for biodiversity, climate, and community benefits. The Darkwoods Forest Carbon Project was developed and Validated and Verified to the Verified Carbon Standard (VCS), AFOLU Methodology VM0012 as a 100 year Improved Forest Management (IFM) – Logged to Protected Forest (LtPF) carbon project on April 21, 2011. The project was Validated and Verified to the Climate, Community, and Biodiversity Standard (CCB) on December 15, 2016 and February 27, 2018, respectively.

Prior to the project, the Darkwoods property was managed for timber production as a private timberland, and was put up for sale under a valuation based on timber and other development values. The Darkwoods Forest Carbon Project involves the acquisition and perpetual management of the property for long term conservation and improvement of ecological function and values through the protection of the forest and various conservation-based forest and land management activities.

The primary objectives of the Darkwoods project is to actively manage and protect the property to achieve climate benefits, biodiversity benefits, water benefits, and other ecosystem services benefits; while retaining substantial net benefits for local communities.

The Darkwoods project achieves net GHG emission reductions through the avoidance of emissions due to conventional logging in the baseline scenario, along with additional carbon sequestration by retaining additional forest biomass and older forests across the project area over time.

The project achieves an exceptional level of biodiversity protection and enhancement through the protection and management of a regionally significant land area with fully functional natural ecosystems and critical habitat for key endangered species.

The project achieves a significant level of water benefits that improve water quality, quantity, timing of flows, and important aquatic habitat improvements over the baselines conditions through the retention and management of riparian habitat, watershed-level forests and ecosystems, and road and access management.

The project area has no communities within or dependent on the project area, however the project achieves net socioeconomic benefits for nearby local communities through limited forest and conservation management and research activities, ecosystem services provisioning, recreational opportunities, and improving community amenity values. The management of the property for these purposes does not generate material levels of net revenue, and the sale of carbon offsets and other ecosystem services benefits are critical to funding significant ongoing property conservation management and ownership costs.

The major objectives of the project are to protect and enhance the long-term function of the natural ecosystems on the Darkwoods property, including:

1. Protecting and enhancing biodiversity through the conservation and management of fully functioning natural ecosystems and key species habitat over time, including:
    a. Restore and maintain Mountain Caribou habitat and movement areas.
    b. Restore and maintain hydro-riparian ecosystem integrity, riparian forest structure and function, fish, and hydrologic functioning.

c. Restore and maintain stand structure within the range of natural variability for Dry ICH forest.

d. Restrict human access to core Mountain Caribou and Grizzly Bear habitat.

e. Maintain or restore Grizzly Bear habitat and movement areas.

f. Restore and maintain old-forest attributes in old-growth and recruiting cedar-hemlock forests.

g. Maintain the integrity of rare ecosystems and habitat features.

h. Restore and maintain full suite of naturally occurring species and ecosystems.

i. Control noxious weeds and other problematic invasive species.

2. Protecting current carbon stocks and future carbon sequestration potential of the ecosystems across the property, including:

   a. Reducing timber harvesting to low levels related to long term ecosystem and habitat management, natural disturbance risk reduction, salvage, or other conservation based objectives

   b. Monitoring property carbon stocks over time

3. Protecting fully functional ecosystems that provide important services and amenities to local communities, while providing diverse economic opportunities; including:

   a. Providing ongoing direct economic opportunities through project activities, expenditures, and operations;

   b. Protecting the property to retain or enhance ecosystem services provisioning, retaining visual quality and other features contributing to regional wildland amenity values, and allowing limited low impact backcountry recreational and other use opportunities as appropriate

   c. Supporting research, monitoring, and other diverse economic opportunities across the property, as feasible

2.1.2 Description of the Project Activity

Project Activity 1 – Conservation of current forests and ecosystems:

The baseline scenario involves timber harvesting to maximize economic opportunities and achieve standard industry returns on investment; which results in a baseline scenario of harvesting ~300,000 m^3/year for the first 15-20 years, followed by periodic harvesting as new stands mature to harvest age over the project lifespan. The primary objective and activity of the project is to avoid this timber harvesting regime and manage the property for conservation purposes. This reduction in timber harvesting results in:

1. Climate impacts: avoidance of GHG emissions resulting from the high level of timber harvesting in the baseline scenario over the project timeline

2. Community impacts: protection of key ecosystem services provisioning related to water quality, air quality, biodiversity, etc. Retention of regional wild land related amenities.

3. Biodiversity impacts: avoidance of habitat modification and operational impacts of logging in the baseline scenario. Retention of large-scale areas of fully functional natural ecosystems and habitat.

Project Activity 2 – Conservation-based forest management:

NCC intends to undertake various conservation management activities, including low levels of timber harvesting, road maintenance and deactivation, silviculture, and various other property management activities to achieve and enhance long term conservation management objectives. The largest impact on forest conditions and GHG emissions during the project will result from an expected low level of timber harvest of approximately 10,000 m^3/year.

1. Climate impacts: avoidance of GHG emissions related to larger scale timber harvesting in the baseline scenario over the project timeline
2. Community impacts: protection of key ecosystem services provisioning related to water quality, air quality, biodiversity, etc. Retention of regional wild land related amenities. Retain and provide a set of limited direct economic opportunities for forest products and forest management in the local communities
3. Biodiversity impacts: avoidance of habitat modification and operational impacts of industrial logging in the baseline scenario, including road deactivation. Retention of large-scale areas of fully functional natural ecosystems and habitat.

Project Activity 3 – Project monitoring.

The project undertakes a variety of ongoing forest and ecosystem inventory activities to monitor property management objectives. In addition to maintaining a comprehensive forest GIS inventory system, NCC has installed an network of permanent plots to measure the current and ongoing carbon balance of the Darkwoods forests over time.

1. Climate impacts: the inventory activities have no material impact on climate other than as related to successfully implementing the carbon project itself.
2. Community impacts: monitoring activities provide direct economic opportunities to local and regional people.
3. Biodiversity impacts: the activity has no material impact on biodiversity other than as related to successfully implementing the project itself.

Project Activity 4 – Biodiversity project monitoring.

NCC has designed and implemented a robust conservation plan and monitoring program for Darkwoods. The Darkwoods property participates in a series of scientific research projects related to key habitat, mapping, threatened species, and other elements.

1. Climate impacts: the activity has no material impact on climate other than as related to successfully implementing conservation plans on Darkwoods.
2. Community impacts: monitoring activities provide direct economic opportunities to local and regional people. As part of the results of conservation planning, NCC has implemented a public access permitting system, along with a series of user lease agreements and other access planning processes that provide opportunities for community members to interact with the Darkwoods property.
3. Biodiversity impacts: the activity has no material impact on biodiversity other than as related to successfully implementing the conservation management of Darkwoods itself.

2.1.3 Implementation Schedule

Date	Milestone(s) in the Project's Development and Implementation
Apr. 1, 2008	Project start date - Darkwoods property acquisition closing date and start of project activities.
Apr. 21, 2011	VCS project validation

Apr. 21, 2011	VCS project verification for periods 2008, 2009, 2010
Mar. 25, 2014	VCS project verification for periods 2011, 2012
Dec. 15, 2016	CCB project validation
Feb. 27, 2018	VCS project verification for periods 2013, 2014, 2015, 2016
Feb. 27, 2018	CCB project verification for periods 2013, 2014, 2015, 2016
Dec., 2019	Expected VCS & CCB verification for periods 2017, 2018
Jan., 2020	Expected VCS & CCB verification for period 2019
2019	SD VISta project validation undertaken
Annually/ongoing	Project implementation, property management activities
Annually/ongoing	Project area monitoring

2.1.4 Project Proponent

Organization Name	Nature Conservancy of Canada (NCC)
Role in the Project	Project Proponent
Contact Person	Rob Wilson
Title	Director - Carbon Finance & New Conservation Strategies
Address	Suite 410 – 245 Eglinton Avenue East, Toronto, Ontario, Canada M4P3J1
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2.1.5 Other Entities Involved in the Project

Organization Name	RainCloud Forests
Role in the Project	Project Developer
Contact Person	Mike Vitt
Title	CEO
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2.1.6 Project Type

AFOLU Improved Forest Management (IFM) – Logged to Protected Forest (LtPF)

This is not a grouped project.

Project Start Date: April 1, 2008

Project Lifetime: 100 years

GHG Accounting Period: 100 years

Previously applied and validated project certifications:

Methodology: VM0012 - Improved Forest Management in Temperate and Boreal Forests (LtPF), v1.0

Tool Applied: VCS AFOLU Non-Permanence Risk Tool v.3.3

CCB Validation Standard: CCB Standards Edition 2
Validated Project Description Documents (see all documentation on the VCS Project Database: https://www.vcsprojectdatabase.org/#/project_details/607):

- Verified Carbon Standard (VCS) PDD: Darkwoods Forest Carbon Project – Version 1.8. Validated April 21, 2011. Project ID: 607.

- Climate, Community and Biodiversity Standard (CCB) PDD: Darkwoods Forest Carbon Project – Version 1.8. Validated June 30, 2014. Project ID: 607.

2.1.7 Project Location

The Darkwoods property is a 54,792 ha (135,394 acre) contiguous parcel of fee simple private property in south eastern British Columbia just north east of the municipality of Creston. The Darkwoods property is bounded by Kootenay Lake on the east and various crown and private land on the other property boundaries. A significant in-holding property in the center of the Darkwoods property has been recently acquired by NCC and the Province of British Columbia (closed 2019) with management plans under development that should be similar/compatible with the project management activities on Darkwoods.. that is owned and managed by the Wynndel Box and Lumber Company, with access rights through the Darkwoods property. The property and project boundaries are shown in Figure 1, and Figure 2.

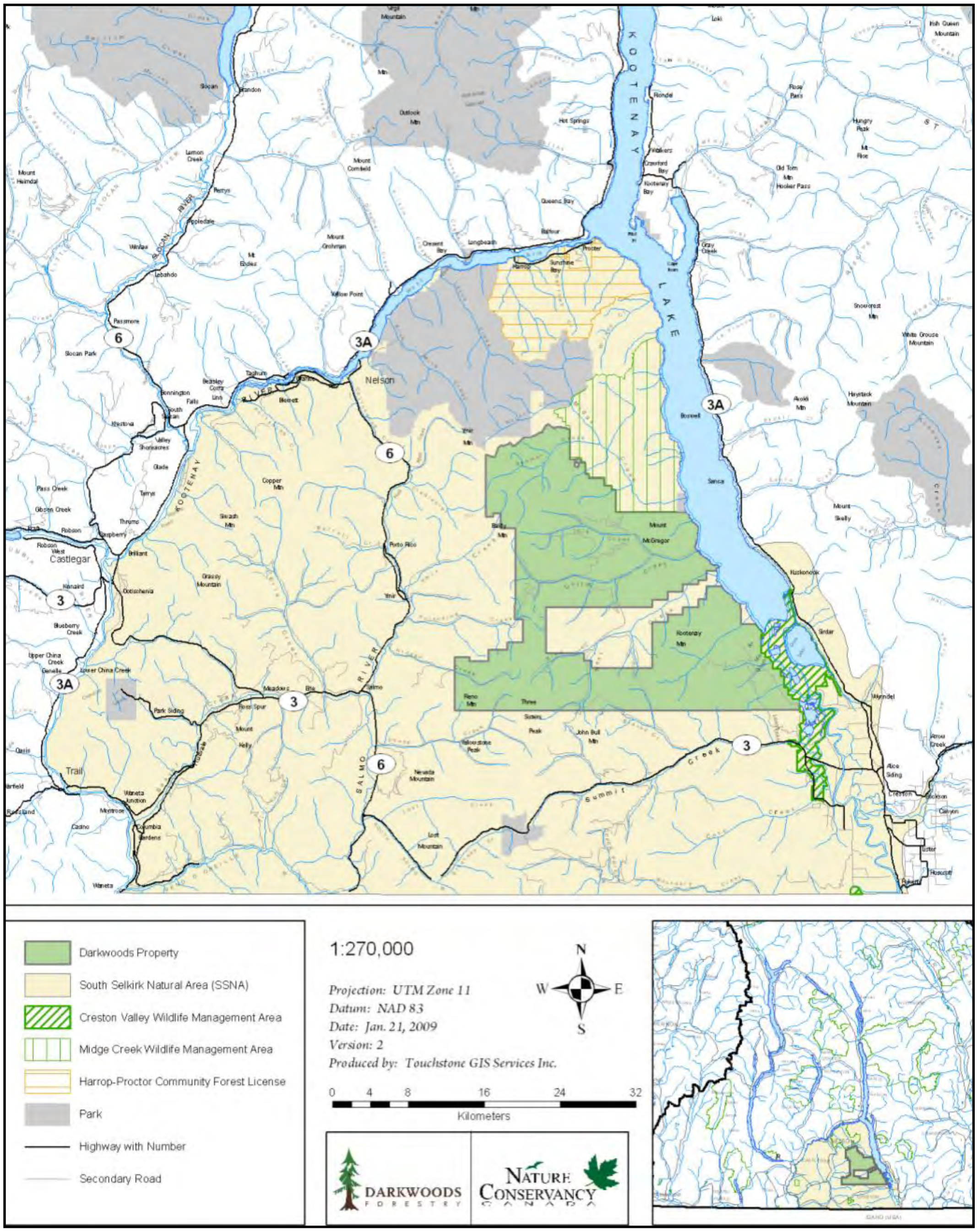


Figure 1. Darkwoods Property Boundary and Project Area Overview Map

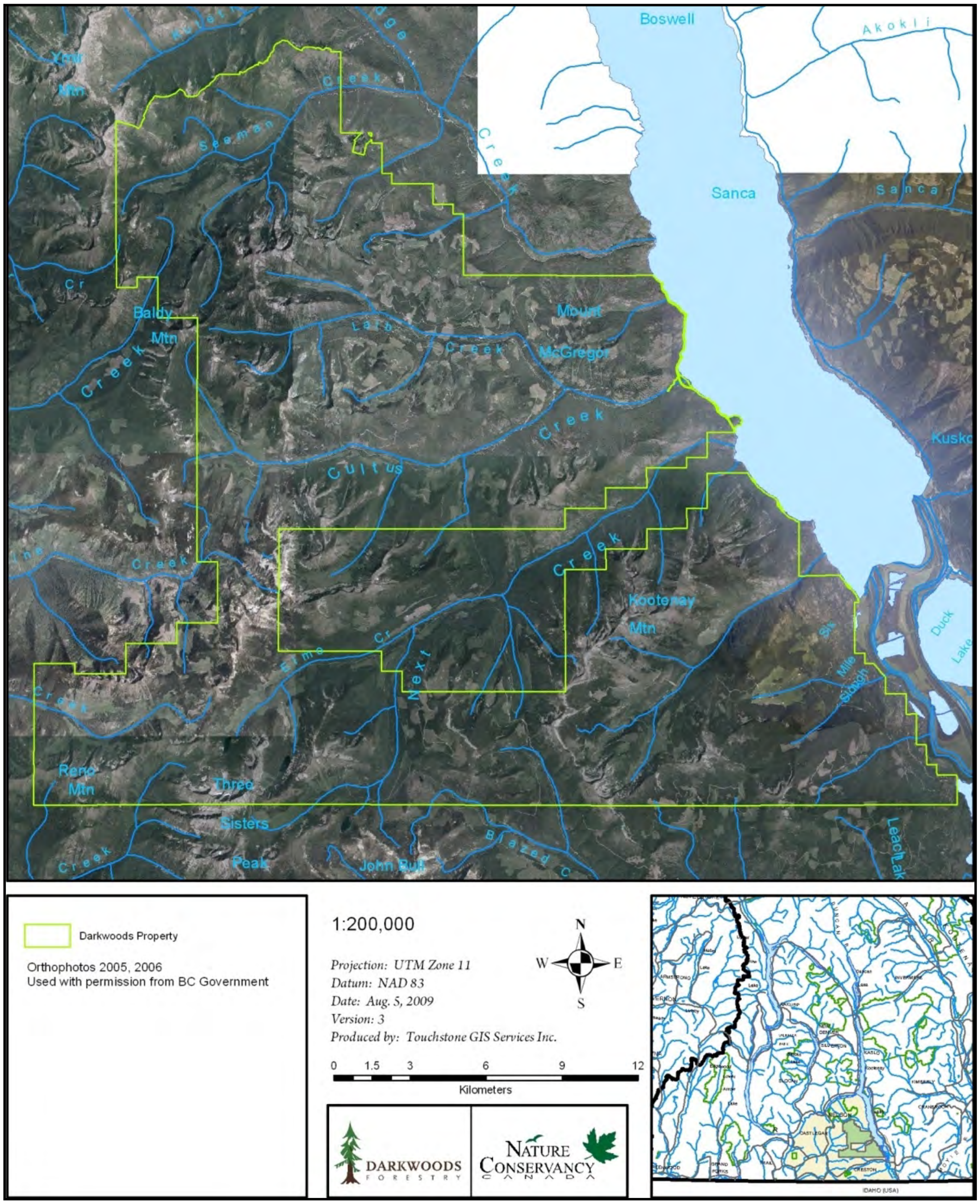


Figure 2. Darkwoods Property Ortho Photo Mosaic

2.1.8 Baseline Scenario

The Darkwoods project area is a private fee simple property. The property is located near the southern extent of the Selkirk Mountain Range, and within the British Columbia Kootenay Lake Timber Supply Area in South Eastern B.C. The property is generally within the area commonly known as the interior wet belt (an inland temperate rainforest area) and contains variable mountainous terrain and bioclimatic conditions. Further detailed property area biogeoclimatic area descriptions can be found in the project's VCS PD document.

Prior to the start of the project, the property was owned and managed for timber production by a number of private interests since the land was granted in 1897. The previous owners, the Pluto Darkwoods Corporation, managed the property for over 40 years, and had put the property up for sale in a bid process in the period prior to 2008. This sale was based on land valuations by independent certified appraisals based on the timber harvesting and potential other development opportunities on the property[1].

The most plausible baseline scenario was evaluated and selected following the processes outlined in the VCS requirements and the VCS methodology VM0012, and is based on the independent timber valuation appraisal upon which the property value and acquisition terms were based. This appraisal projects an economic timber harvest schedule to remove mature and excess timber inventory following legal requirements and common practices for private managed timberland in British Columbia, and typical of private timberland management in the region. This baseline harvest results in the logging of the available mature timber over a period of 15-20 years to provide return on investment for the alternative property buyer, followed by periodic harvests as forest stands reach economic maturity over the project timeline.

As modeled, the baseline scenario would harvest approximately 300,000 m^3 per year for 15-20 years, and then revert to a variable harvest level averaging 59,000 m^3 per year (ranging from 0 to 165,000 m^3 per year) until year 100.

Additional details and data related to the assessment and selection of the baseline scenario, and the methods used to model the baseline scenario can be found in the project's VCS PD documentation.

The baseline scenario results in substantial GHG emissions and negative impacts on biodiversity, water, and ecosystem services due to the removal of forest biomass during timber harvesting and related activities. Although the timber harvesting in the baseline generates short term and periodic economic gains for certain segments of the local community, the baseline results in the significant loss of other benefits to the local community[2].

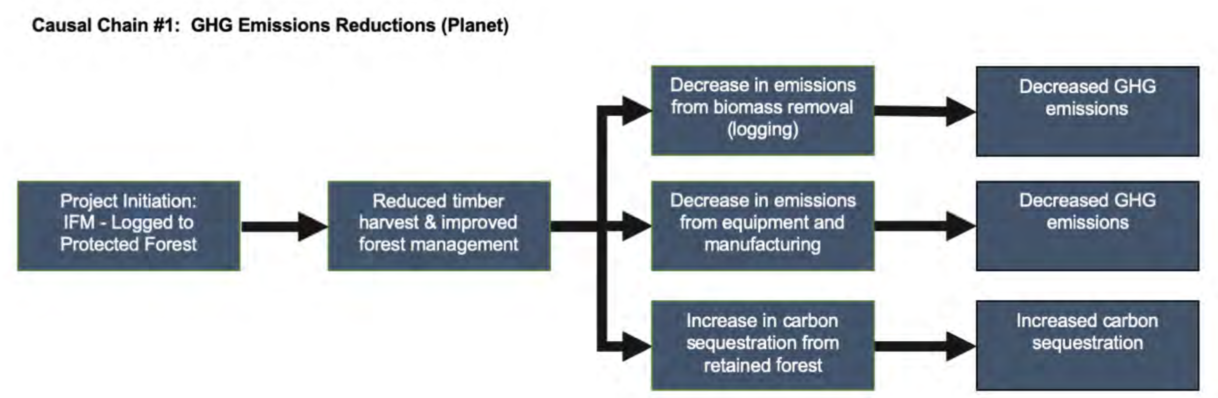
2.1.9 Causal Chain(s)

The primary impact of the implementation of the project is to avoid the carbon, biodiversity, and community net impacts associated with common practice timber harvesting in the baseline scenario. The project, in comparison, retains and manages forests to develop and improve fully functional ecosystems.

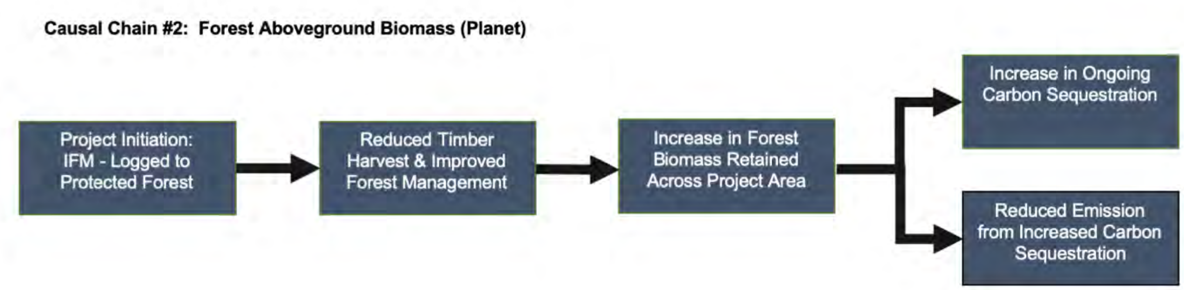
[1] Additional details of the history of the property, conditions prior to the project, additionality assessments, and the baseline scenarios evaluated and selected as most plausible are available in the project VCS and CCB Project Description documents.

[2] Additional details and description of the impacts of the baseline on carbon, biodiversity and community are available in the project CCB Project Description document.

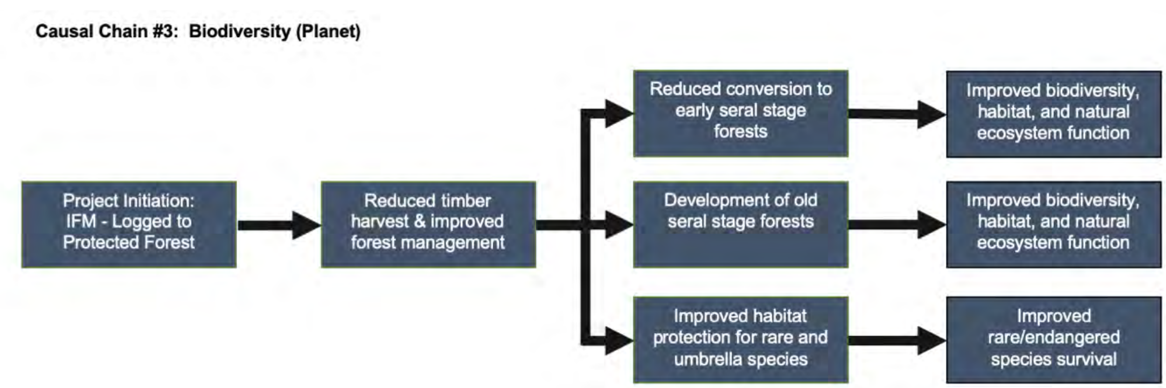
Causal Chain 1: GHG Emissions Reductions (Planet)



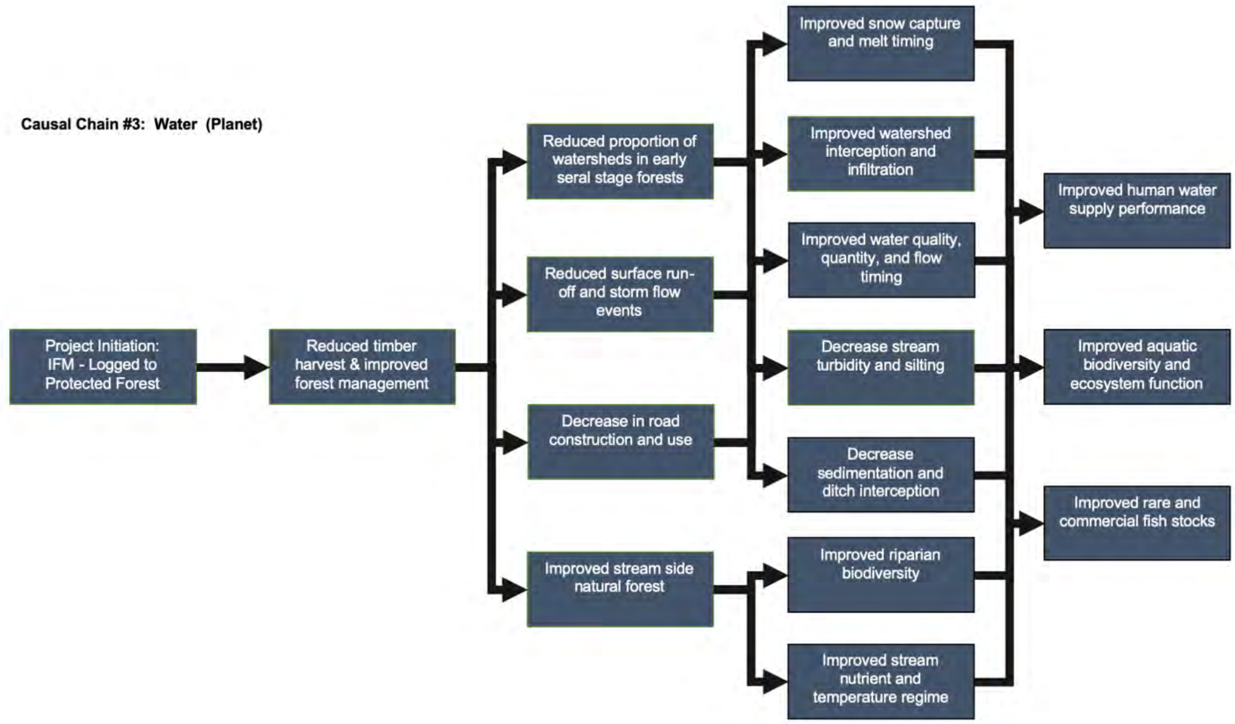
Causal Chain 2: Forest Aboveground Biomass (Planet)



Causal Chain 3: Biodiversity (Planet)

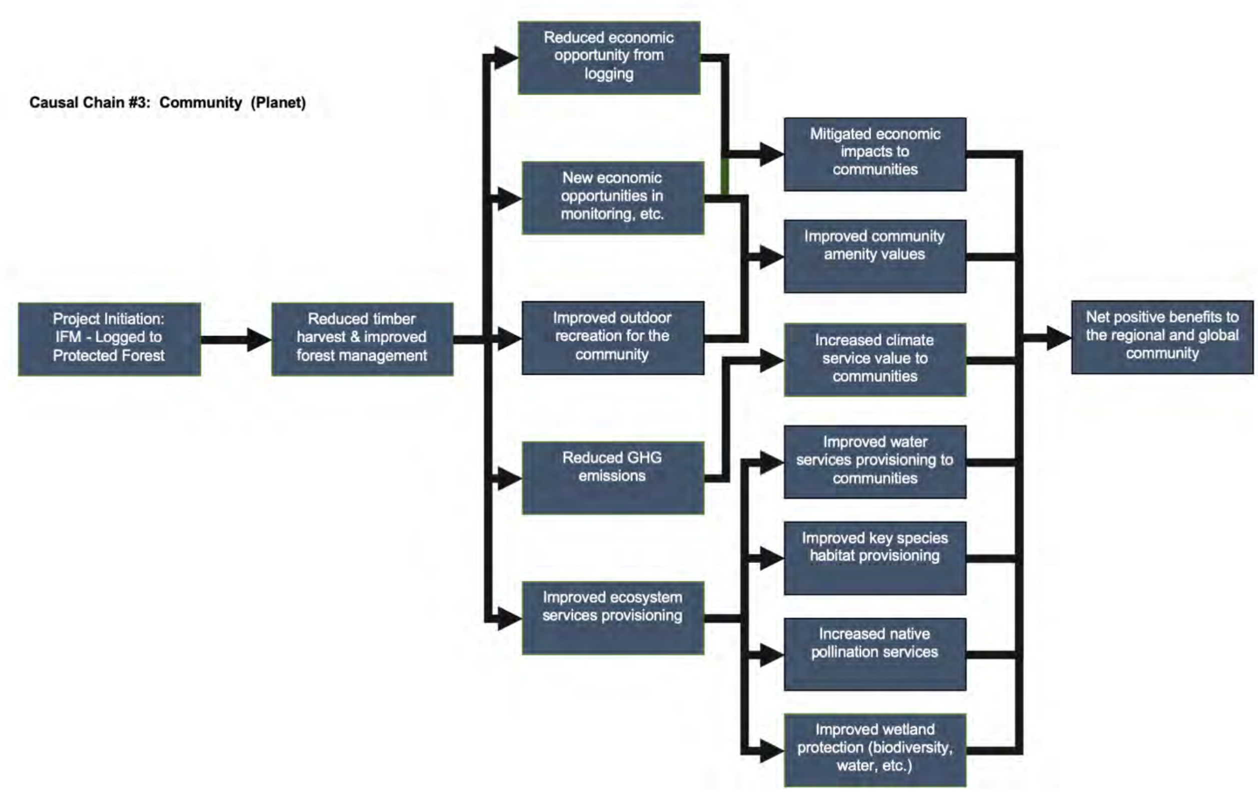


Causal Chain 4: Water (Planet)[3]



[3] A review of the water quality impact of forest harvesting and roads: https://www.for.gov.bc.ca/hfd/pubs/docs/lmh/Lmh66/Lmh66_ch12.pdf and https://soils.ifas.ufl.edu/media/soilsifasufledu/sws-main-site/pdf/technical-papers/Orndorff_Adam_Immediate_Release.pdf. A review of Equivalent Clearcut Area (ECA) as a measure of watershed impacts: https://www.for.gov.bc.ca/hfd/pubs/docs/en/EN118.pdf

Causal Chain 5: Net Community Benefits (People and their Prosperity)[4]



2.1.10 Threats to the Project

The risks and threats to the sustainable development are outlined in detail in the VCS Non-Permanence Risk Assessment documentation. There are no additional threats to the project specifically related to the SD VISta elements in the project.

Summarized, the project area generally has the spatial scale and biological, terrain, and climatic diversity to be low risk for significant loss to carbon, biodiversity, and community values in the project. Forest fires do occur periodically on the property and are likely one of the higher direct risks to climate and biodiversity impacts. However, fires are rare in this wet/moist bioclimatic zone, while the steep and diverse terrain of Darkwoods makes large scale fire damage rare. NCC has a staffed field office and constant presence on the property, and has a fire fighting agreement with the British Columbia Ministry of Forests who have significant firefighting assets in nearby Creston, B.C. Collectively these features strongly reduces the risk of fires and significant damage to forest biomass on the property.

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[4] The project has not made claims related to SDG indicators for People and their Prosperity primarily because the UN SDG indicators are not clearly targeted at the developed country conditions and local community situation found in the project region. However, the project can demonstrate a net community benefit that is an important element to the project and local stakeholders. Further details and supporting information related to the assertion of no net negative impact to the local communities can be found in the project's validated and verified CCB PD.

Human induced threats are also rare on the Darkwoods property due to its relatively remote location, gated road access controls, and NCC's recreational use permitting system; along with being located in a fully developed region with excellent judicial and enforcement systems. The property is owned by the largest conservation group in Canada (a not-for-profit organization) and is protected by perpetual conservation easements and the Canadian Federal Eco-gifts Program participation agreement, which collectively provide excellent ownership risk mitigation.

2.1.11 Benefit Permanence

The Darkwoods property is owned and managed for conservation purposes in perpetuity by the Nature Conservancy of Canada. The property also participates in the Canadian Eco-Gifts Program that provides legally binding perpetual permanence protection and enforcement mechanisms to insure the property is managed for conservation purposes in perpetuity.

The result is the benefits to climate, community, and biodiversity will be retained in perpetuity as the property retains, regrows, and enhances fully functional ecosystems across the property.

2.2 Stakeholder Engagement

2.2.1 Stakeholder Identification

The Darkwoods project area is the private, fee simple property (Figure 4) that sits within a economically developed and wealthy geography in British Columbia, where the socio-economic conditions in the general region around the project area are stable and diversified, with very well developed infrastructure linking an integrated economy throughout the southeastern B.C. region and generally across the province and Canada. Internet and phone access are nearly ubiquitous and highly used across the region. There are no communities, indigenous people or other public communities or ongoing public use for income, livelihood, or critical cultural values by any people living or dependent on resources within the project area that would be affected by project activities. Access roads onto the Darkwoods property are gated and public access controlled by NCC. The projects validated CCB Project Description document contains more detailed descriptions of the surrounding communities and regional economic conditions.

Therefore, there are no stakeholders whose livelihoods are directly affected by the project on the project area that require a specific or complex stakeholder consultation or involvement process.

NCC has, however, endeavored to communicate information related to the property acquisition and various project and property activities and impacts with a wider range of interested stakeholders in the local community who may be indirectly impacted by the project or who are generally interested in the property area, project, and/or NCC initiatives and impacts in the area. It should be noted, however, that although the local community are certainly interested in Darkwoods, there is a limited “attention span” in these modern communities for lengthy or constant communication from NCC or in relation to the project. Therefore NCC takes a strategic approach to communicating by providing information via the website and social media for any interested stakeholder to access on a continuous basis, and then more targeted outreach to meet with specific groups on an ‘as-needed’ or ‘as-requested’ basis; while limiting more general community outreach and public meetings to specific opportunities when they can garner optimal interest and attention on a periodic basis.

Although there are no stakeholders directly on the project area, these local community communication processes have allowed NCC to identify “interested stakeholders” and groups potentially interested in specific elements of the Darkwoods project, while also allowing any other interested person's to communicate directly with NCC as needed.

2.2.2 Stakeholder Description

As noted in Section 2.2.1, there are no specific stakeholders that live on or rely on the project area for their livelihood, and the project has not identified any stakeholders directly affected by the project or project activities. Therefore, the focus in the project is on identifying and communicating with interested stakeholders and the local community generally.

NCC has undertaken a significant and ongoing effort to communicate with the local community generally on a periodic basis, and specifically with any directly interested stakeholders on an as-needed, or at-request basis. Additionally NCC has an active permanent presence in the local community and region, along with active webpages and social media accounts, such that any other interested stakeholder has the opportunity to communicate with NCC through direct contact and in periodic public meetings.

There are a few examples of local community members who are active on the project area, but who are not negatively impacted by the project activities, including:

- There are currently 27 non-permanent recreational cabins/trailers with associated lake moorages located on a small area on the eastern edge of the property along Kootenay Lake, known as Tye. NCC has permitted a continuation of the pre-existing seasonal recreational cabin usage through a process of annual rental agreements. NCC retains a year-round caretaker to oversee and care for this Tye recreation area. NCC communicates directly with the Tye community on an as-needed/ongoing basis.
- There are a few (1-2 active) small (3-4 workers) exploration and mining/prospecting operators occasionally active on the property (but no productive mines). These operations have a minimal surface impact (<1 hectare) and their impacts on carbon emissions are immaterial. NCC communicates directly with these operators on an as-needed basis.

In general, the local interested stakeholders that actively engage with NCC related to the Darkwoods property and that may be directly or indirectly impacted by the project include:

- Project-based contracting opportunities for local road construction, maintenance, timber harvesting, and/or silvicultural activities within the project
- Project based forest management consulting for inventory, planning, monitoring, or other project related work
- Backcountry recreational users
- Researchers undertaking scientific studies

NCC communicates periodically with these interested stakeholders as necessary or requested on an annual or periodic basis.

In addition, there are regional groups, associations and members of the public in Nelson, Creston, and the surrounding areas that occasionally express interest in conservation initiatives, research, and other general happenings on the Darkwoods property and the surrounding area. NCC meets and communicates with these groups on a periodic basis or upon request related to the carbon project, project activities, and plans on Darkwoods; along with other regional NCC activities.

The property sits within the traditional territory of the Ktunaxa First Nation. Generally private property rights in Canada are respected by First Nations, who primarily focus on land use rights and territorial issues on the larger Crown land controlled by the province. The Ktunaxa are important regional stakeholders who have an interest in cultural and environmental considerations on all areas of their territory, including Darkwoods; however, there is no material level of ongoing use, livelihood activities, or special cultural values on the project area and within the project zone that are materially impacted by the project activities (i.e. Choquette, 2010). NCC has a strong and active working relationship with the Ktunaxa and is engaged on several regional initiatives of mutual interest.

2.2.3 Stakeholder Consultation

The Darkwoods project relates to a private acquisition of a fee-simple property with no people living on or relying upon the project area for their livelihood. Hence, no direct consultation process was necessary. However, NCC undertook an extensive outreach and public/interested stakeholder communication process in the local communities in the months leading up to and after the acquisition of Darkwoods (i.e. the project start date). Further details and examples from these initial stage engagements are presented in the project's VCS PDD.

On an ongoing basis the primary method of communication with interested stakeholders for the project is the NCC and Darkwoods webpages, along with NCC's social media presence. The local and regional communities have full access to modern communication and internet connections, and therefore NCC primarily uses the website and social media software platforms for general and/or broad communication about the project and related activities and key contacts within NCC. In addition to direct phone and email access to local NCC staff, many of the social media platforms maintained by NCC are designed for two-way communication such that any interested stakeholders can communicate with NCC about the project area as needed.

NCC also undertakes periodic ongoing engagements with interested local stakeholders annual on an as-needed or as-requested basis, as generally noted in Sections 2.2.2 and 2.2.3. These engagements involve direct meetings, public sessions, and responses to individual group or persons.

Examples of interested stakeholder identification and engagement during the project start-up phase (2008-2010) are shown in Table 2.

Table 2. Darkwoods Interested Stakeholders Meetings at Project Initiation

Timeline	Stakeholders Met	Comments
Aug 2008	Technical Advisory Committee Members Outdoor clubs Regional District of Central Kootenay (RDCK) Ministry of Tourism, Ministry of Mines, Ministry of Environment Ministry of Forest and Range	
Sept 2008	Regional District of Central Kootenay (Sept. 20) Rod and Gun Clubs of Creston, Trail, Nelson and the West Arm Local ENGO's and researchers Snowmobile Clubs of Creston and Fruitvale Mining Association of BC Nelson Chamber of Mines	
Sept 2008	Open House Information Sessions: September 22, 2008 Creston – 65 people attended Sept. 23/08 in Salmo – 35 people attended Sept 24/08 in Nelson – 45 people attended	Advertised in the local media and through the use of Public Service Announcements.. Direct invites to relevant partners, stakeholder, and regional NCC donors Open house format: open meet and greet. NCC display materials, Darkwoods maps and hand-outs. NCC people at each event. Guests were also encouraged to fill in a comment form
Oct 2008	Public meeting sponsored by Wildsight (local NGO) in Creston - 60 people attending Technical Advisory Team meeting Association of British Columbia Snowmobile Clubs (ABCSC) Ministry of Environment (MoE) Fish and Wildlife Compensation Program (FWCP) Partnership opportunity meeting with Selkirk College Sinixt Nation Meteorological meeting with Alistar Frasier Ministry of Forest and Range on a number of issues:	
Apr 2009	BC Hydro's Fish and Wildlife Compensation Program (FWCP), Selkirk College, MoE, MoF, BC Tourism, BC Integrated Land Management Bureau (ILMB) and RDCK	
Dec 2009	Selkirk College. MoE, MoF, Ministry of Mines, BC Tourism, ILMB and FWCP BC Wildlife Federation	
July 2010	Met with Tye residents Ktunaxa Nation Council Mineral tenure holders and the government agent RCMP	

Sept 2010	Open Houses in Creston, Nelson, Salmo	

Examples of interested stakeholder engagement from the most recent verified VCS/CCB monitoring period (2013-16) are shown in Table 3 and Table 4.

Table 3. Darkwoods public access permit monitoring results

Darkwoods Public Access Permits	Year	# Permits	Est. Non-Permitted Use
Public Access	2013	172	150
Public Access	2014	98	150
Public Access	2015	80	150
Public Access	2016	138	150

Table 4. Recreational groups engaged with NCC during the previous monitoring period

Recreational Groups	Years
Kootenay Mountaineering Club	2013-2016
Beaver Mountain Snowmobile Association	2013-2016
Kokanee Snowmobile Club	2013-2016
Nelson Snowgoers Club	2013-2016
Creston Rod and Gun Club	2013-2016
Association of BC Snowmobile Clubs	2013-2016
West Kootenay Naturalist Club	2013
West Kootenay Camera Club	2013
Selkirk Rock and Mineral Club	2015
Western Canada Bryophyte and Lichen Interest Group	2014
Nelson Camera and Naturalist Club	2013
Climbers Association	2013

Table 5. Scientific researchers engaged on Darkwoods over the past ~5 year

Scientific Researchers	Topic	Years
Michael Proctor	Grizzly Bears	2013-2016
Adrian Leslie	Whitebark pine	2013-2014
Marc-Andre Beaucher	Nighthawks	2015
Grant McHutchen	Grizzly Bears	2013-2016

Masse Environmental Consultants	Bull Trout Redd Surveys	2015
Stefan Himmer	Huckleberries	2015
Gerry Nellestijn	Bull Trout	2016
Olivia Lee	Bryophytes and Lichen	2014
Leo DeGroot	Caribou	2013-2016
Highlander Forestry	FLNRO subcontractors	2014
Cori Lauson	Bats	2014
Jeremy deWaard	Anthropods	2014
Pam Dykstra	FLNRO subcontractors	2013
Olivia Lee	Botany	2014
Steve Ardnt	Kokanee and Bull Trout	2013-2016
F&W Compensation Program	Bull Trout Redd Counts	2017-2019
BC Conservation Data Centre	Red and blue listed species	2016
BC MOE	Caribou camera research project	2018-2019
Masse Environmental	Mycology inventory	2017-2019
Selkirk College	Whitebark Pine remote sensing	2019
Transborder Grizzly Bear Project	Grizzly & Huckleberry research	2017-2019
Living Lakes Canada	Water quality monitoring sites	2019
C.Kootenay Invasive Sp. Society	Longterm invasive weeds monitoring	2018-2019
Simon Fraser University (MS Thesis)	Caribou habitat restoration	2018
BC Conservation Data Center	Plants at risk inventory	2017

2.2.4 Continued Consultation and Adaptive Management

As noted in Section 2.2.5, NCC has an ongoing program for outreach and communication with local and other interested stakeholders in the Darkwoods project area. In general, people within the local communities utilize the website and social media for information and updates from NCC about Darkwoods on an ongoing basis. NCC meets with various local community groups and stakeholders on a periodic basis, generally when requested or otherwise when information of sufficient interest to those stakeholders arises. NCC also undertakes general public meetings when there are new events or sufficient interest from the community. And finally, NCC maintains and office and permanent staff near the project area, which stakeholders can contact directly by phone or email (or via the website and social media) for information or any concerns. Collectively, these communication process provide a mechanism for ongoing consultation with interested stakeholders and the local community generally. NCC gathers and shares input from community members and meetings internally, and incorporates feedback adaptively in property management plans and annual operating plans and policies when relevant and feasible. NCC tracks and reports on these ongoing stakeholder and local community activities during project verification processes.

2.2.5 Anti-Discrimination

Canada and British Columbia maintain a robust set of laws and regulations that are covered under various federal and provincial human rights and labour legislation. The Nature Conservancy of Canada is a prominent national organization that is compliant with the laws of Canada and B.C.. NCC also has well-developed internal human resources policies governing the management and conduct of NCC's employees: The Nature Conservancy of Canada Management Policy: Human Resources Policy (2012) and the NCC Darkwoods Occupational Health & Safety Program (2012). These policies include statements related to discrimination and related reporting and management practices. These policies are reviewed and updated by senior management annually, and all employees are trained and required to comply with these policies.

2.2.6 Worker Training

NCC's policies related to health and safety, including safe operating procedures, are fully documented in: NCC Darkwoods Occupational Health & Safety Program (2012). Employees and contractors maintain programs that meet or exceed the B.C. Forest Safety Council requirements and WorkSafe B.C. Regulations. Other specific technical tasks such as carbon monitoring plot installation and measurement are trained at the start of each field season using standard operating procedure documentation.

2.2.7 Equal Work Opportunities

See Section 2.2.5.

2.2.8 Workers' Rights

Canada has very well developed labour legislation governing workers' rights. Responsibility for the establishment of labour legislation generally falls under provincial jurisdiction, along with federal authority in certain industries. This legislation governs minimum employment standards that by law define and guarantee workers' rights within the workplace. Each province, as well as the federal government, has legislation governing workplace health and safety, as all Canadian workers have the right to work in a healthy and safe environment. Human rights provisions, fair employment practices and equal pay and anti-discrimination laws are covered under federal and provincial human rights and labour legislation.

In British Columbia, the B.C. Ministry of Labour, Employment Standards Branch is responsible for the administration of the provincial Employment Standards Act. This Act covers the following topic areas: Minimum Wage, Minimum Daily Pay, Meal Breaks, Paydays and Payroll Records, Overtime, Averaging Agreements, Uniforms & Special Clothing, Deductions, Statutory Holidays, Employing Young People Under 15, Compensation for Length of Service, Annual Vacation, Vacation Pay, Leave From Work, Temporary Foreign Workers, Collective Agreements and Resolving Disputes.

The Nature Conservancy of Canada has well-developed internal human resources policies governing the management of NCC's employees. The overarching Human Resources Policy is reviewed annually by senior NCC management and updated when deemed necessary. All NCC employees are informed of the existence of this policy on NCC's intranet website, which addresses the standards, requirements and procedures governing an employee's employment by NCC. All new employees are required to sign a disclosure indicating agreement and adherence to these policies and all NCC employees must renew this agreement on an annual basis.

Subcontractors, defined as service providers who bill NCC for their services, must sign an Agreement for Services contract, the standard template of which includes a separate provision entitled Compliance with Laws and Policies, as follows: "The Contractor shall be knowledgeable of and comply with the provisions of all applicable legislation, By-Laws of local authorities and Regulations in the Province of British Columbia and shall specifically ensure that the Contractor and his/her/its employees are knowledgeable of and performs all obligations imposed by the *Occupational Health and Safety Act* or equivalent in the Province of British Columbia".

In summary, NCC is a national organization fully compliant with the laws and regulations of BC and Canada. NCC employment policies and worker guidelines are documented in the latest versions of: The Nature Conservancy of Canada Management Policy: Human Resources Policy and the NCC Darkwoods Occupational Health & Safety Program.

2.2.9 Occupational Safety Assessment

Darkwoods project activities are typical operations for forestry operations and timberland in BC, and include typical forest operations (logging, road work, etc.) and various forest field work. There are no unusual or unique activities, and therefore risks are typical of operations undertaken commonly across the forest industry.

In Canada, all 3rd party contractors operate as "independent contractors", a specific legal term that requires the contractors to operate and direct their delivery of services and employees independently. Therefore, operational safety procedures are the purview of each contractor. The WorkSafe BC regulations and BC Forest Safety Council SAFE Company programs are the primary methods in BC to assure worker safety in the forest industry, and include training and procedural materials used across the industry. Various levels of performance are self-audited by these programs.

NCC's policies related to health and safety, including safe operating procedures, are fully documented in: NCC Darkwoods Occupational Health & Safety Program. As per this document and other NCC annual plans, employees undergo various health and safety awareness and training programs, as applicable to their job function.

NCC contracts mandate that all contractors have a valid and active WorkSafe BC number. The contractor is then responsible to WorkSafe BC (https://www.worksafebc.com/en/about-us/who-we-are/mission-vision-values) to ensure that legal regulations are followed. Logging contracts also specify contractors are certified as BC Forest Safety Council SAFE Company.

NCC undertakes pre-work reviews with contractors as described in the NCC Darkwoods Occupational Health & Safety Program, which includes a pre-work safety site orientation checklist. NCC also ensures that logging contractors and other large contractors have at least equivalent procedures. Researchers are made aware of safety issues through both a research permit and waiver. And finally, general public access permits are also advised of the risks on the permits document.

2.2.10 Feedback and Grievance Redress Procedure

See Section 2.2.11.

2.2.11 Feedback and Grievance Redress Procedure Accessibility

The Nature Conservancy of Canada is a highly accessible organization and maintains multiple venues for contacting it, including:

- Offices with knowledgeable staff across the region in Nelson, Invermere, Victoria, Vancouver, and Toronto, ON
- Listed phone numbers for each site (phone book and website)
- NCC's website including address, phone number and email contact info for each BC office

- NCC's website's general "contact us" function
- Caretaker at Tye who interacts with Tye leaseholders and visitors and communicates directly to NCC
- Signs posted on the Darkwoods Property with NCC's toll free phone number

When NCC receives an expressed concern from a community member or interested ntestakeholder, NCC deals with it in the most appropriate manner. If the concern is addressed to NCC in writing, NCC will typically respond in writing as soon as possible (examples available upon request). In the more common instances in which NCC receive a phone call or in-person visit, concerns or comments are addressed in person. Regardless of which form of contact has been used, NCC has been able to satisfactorily address outstanding concerns in this manner. If a concern were to be elevated, NCC would engage a third party mediator if warranted or if, in certain instances, NCC is contractually obligated to do so.

NCC has a formal process to receive communications from interested stakeholders at local or regional offices, which are documented locally as received[5]. Any significant or unresolved issues are forwarded to the regional office, where records are kept and follow-up plans developed as necessary.

The B.C. Regional Office with delegated authority from the NCC National Office is the final authority for responding to any issues related to Darkwoods.

NCC has received comments related to Darkwoods throughout the acquisition and project implementation which have been handled through this process, and which have resulted in stakeholder meetings and other resolution actions (examples available upon request). NCC is unaware of any unresolved complaint or conflict material to the implementation of the Darkwoods project.

In general, local community members and interested stakeholders will contact local NCC staff for information or concerns via phone, email or social media.

2.2.12 Stakeholder Access to Project Documentation

Project documentation and updates are made available electronically via the NCC Darkwoods webpage and social media as needed. Interested stakeholders are generally directed to the Verra Project Database website for final project documentation, which it is permanently stored and available to any interested stakeholder. This Verra website documentation includes VCS, CCB, and eventually SD VISta documentation. The public comment period for the draft SD VISta project documentation are made available via the Verra Project Pipeline website and comment process, while NCC provides links and updates for stakeholders via the NCC website and social media accounts. Project documentation is also made available to interested stakeholders by NCC staff by request (this is not common). Generally, NCC and Darkwoods function within the modern electronic world where web and social media contact are the preferred communication methods for local communities and stakeholders.

2.2.13 Information to Stakeholders on Assessment Process

See Section 2.2.12. Additionally, NCC directly contacts key stakeholders by phone or email with notice of potential contact by the project assessor or during a site visit as appropriate.

2.3 Project Management

2.3.1 Avoidance of Corruption

The Nature Conservancy of Canada is a prominent national not-for-profit organization that complies with the laws and regulations of Canada and British Columbia, including those pertaining to corruption, etc. NCC maintains well-developed internal human resources policies governing the management and

[5] As of about 2013 NCC has completed operationalizing its new Land Information System, a robust internal data tracking system, within which calls and emails related to the Darkwoods property are now documented and retained digitially.

conduct of NCC's employees: See The Nature Conservancy of Canada Management Policy: Human Resources Policy (2012), and supporting documentation.

2.3.2 Statutory and Customary Rights

NCC owns the Darkwoods property on a fee-simple basis, and all title and rights documentation is available for review. There are no title or rights disputes on Darkwoods.

2.3.3 Recognition of Property Rights

NCC owns the Darkwoods property on a fee-simple basis, and all title and rights documentation is available for review. There are no title or property right disputes on Darkwoods.

2.3.4 Free, Prior and Informed Consent

No people live on or depend on the Darkwoods project area for their livelihood, and therefore this section is not applicable.

2.3.5 Restitution and/or Compensation for Affected Resources

No people live on or depend on the Darkwoods project area for their livelihood, and therefore this section is not applicable.

2.3.6 Property Rights Removal/Relocation of Property Rights Holders

No people live on or depend on the Darkwoods project area for their livelihood. No removal or relocation of property rights has occurred, and therefore this section is not applicable.

2.3.7 Identification of Illegal Activities

British Columbia has robust and enforced laws and regulations, with strong private property protections overall. This includes a timber stamping/marking and scaling reporting system that makes timber theft difficult and rare. In addition, active provincial wildlife poaching monitoring, reporting, and enforcement systems are in place. Timber theft and wildlife poaching are rare in B.C., and provincial police, forestry, and wildlife enforcement resources are available if any events do occur. There is very low risk of illegal activities materially affecting project carbon stocks or biodiversity. Access routes to Darkwoods are gated and monitored on a regular basis by NCC staff based in nearby Nelson, B.C.

There has been no evidence of material illegal activities on the Darkwoods property to date.

2.3.8 Ongoing Conflicts or Disputes

There are no ongoing conflicts related to rights, lands, territories or resources related to the Darkwoods project area.

2.3.9 National and Local Laws and Regulations

This forest carbon project is designed to be compliant with Canadian and British Columbian laws in both the baseline and forest carbon project scenarios.

The Darkwoods carbon project is focused on conserving forest ecosystems and habitat on private lands, and is inherently compliant with provincial, federal, and international laws and regulations that are targeted at regulating development and non-conservation land use.

The primary set of legislation and regulation relevant to forestry activities on Darkwoods is the Private Managed Forest Land Act (PMFLA), which provides reduced property taxes and local land zoning protection in exchange for compliance with minimum forestry operating standards.

As noted in the VCS PDD, the baseline scenario suggests the implementing entity would opt out of the PMFLA registration and revert to lessor local land zoning minimum requirements. However, the modeled baseline scenario has been conservatively modeled to be fully compliant with (or exceeds) all requirements of the PMFLA.

The Darkwoods property is currently registered by NCC in the PMFLA, and hence the project scenario activities are also compliant with the PMFLA. Generally, the remainder of the legislation outlined below have no or de minimus operational impact on the baseline or project scenario projections.

An overview of the most relevant Provincial and Federal laws and regulations, which might apply in certain circumstances to the Darkwoods property baseline or project scenario (FSC Canada, 2005):

Statute: Private Managed Forest Land Act, S.B.C. 2003, c. 80.

Comments: Sets out the legal framework for forest management on untenured private managed forest land.

Regulations: Private Managed Forest Land Council Matters Regulation, B.C. Reg. 372/2004. Private Managed Forest Land Council Regulation, B.C. Reg. 336/2004. Private Managed Forest Land Regulation, B.C. Reg. 371/2004.

Applicability: This law applies as Darkwoods is a privately managed forest and is registered the PMFLA under NCC ownership.

Statute: Foresters Act, S.B.C. 2003, c. 19.

Comments: Regulates the forestry profession including registration, certification and discipline. Provides for the creation and mandate of the Association of British Columbia Forest Professionals.

Applicability: This law applies as Darkwoods has a forest operation and the Forests Act regulates the forest profession. NCC's Darkwoods operations manager is a B.C. Registered Professional Forester.

Statute: Water Act, R.S.B.C. 1996, c. 483.

Comments: Among other things, applies to persons making changes in or about streams. See provisions in the Water Regulation that apply to forestry operations, particularly 38(2) 41 and 42.

Regulations: Groundwater Protection Regulation, B.C. Reg. 299/2004. Water Regulation, B.C. Reg. 204/88.

Applicability: This act applies to forest operations as it is required that activities around a stream do not negatively impact water quality or wildlife habitat. Generally, the B.C. PMFLA provides suitable compliance with this act and regulations.

Statute: Wildfire Act, S.B.C. 2004, c. 31.

Comments: Regulates the use of open fires in and near forest land. Outlines the government's rights and duties in case of a wildfire.

Regulations: Wildfire Regulation, B.C. Reg. 38/2005.

Applicability: This act applies to government response to wildfire on private lands.

Statute: Fisheries Act, R.S.C. 1985, c. F-14.

Comments: Among other things, provides that no person shall carry on any work or undertaking that results in the harmful alteration, disruption or destruction of fish habitat, and prohibits depositing deleterious substances of any type in water frequented by fish. Provides for authorizations to alter fish habitat.

Regulations: Fishery (General) Regulations, SOR/93-53.

Applicability: This act applies as any work completed on Darkwoods must not result in harmful alteration, disruption or destruction of fish habitat. Generally, the B.C. PMFLA provides suitable compliance with this act and regulations.

Statute: Migratory Birds Convention Act, 1994, S.C. 1994, c. 22.

Regulations: Migratory Birds Regulations, C.R.C., c. 1035.

Migratory Birds Sanctuary Regulations, C.R.C., c. 1036.

Applicability: Darkwoods has several endangered bird species that utilize the property during migration periods. This act protects migratory birds and their nests. Generally, the B.C. PMFLA provides suitable compliance with this act and regulations.

Statute: Navigable Waters Protection Act, R.S.C. 1985, c. N-22.

Comments: Among other things, applies to dumping of fill and to structures or bridges on navigable waters.

Regulations: Navigable Waters Bridges Regulations, C.R.C., c. 1231.

Navigable Waters Works Regulations, C.R.C., c. 1232.

Applicability: Darkwoods borders Kooteney Lake and this act applies to activities that impact these navigable waters as well as potentially the larger sections of small streams/rivers that run through Darkwoods. Generally, compliance with the B.C. PMFLA serves as suitable compliance with these acts and regulations for forestry activities.

International Agreements:

2.3.10 Project Ownership

NCC title documents are properly registered and publicly available through the B.C. Land Title Office. Moreover, all title documents and related plans have been provided to third party auditors and validated/verified as part of the VCS validation/verification of the project. The project area does not contain or involve any areas not owned privately by NCC.

2.3.11 Grouped Projects

The Darkwoods project is not a grouped project.

3 BENEFITS FOR PEOPLE AND PROSPERITY

3.1 Condition of Stakeholders at Project Start

The Darkwoods project area is a private, fee simple, contiguous property with gated road access control. There are no communities, indigenous people or other ongoing public use for income, livelihood, or critical cultural values by any people living or dependent on resources within the project area that would be affected by project activities.

However, there are economic, amenity value, and ecosystem services impacts from implementing the project scenario (in comparison to the baseline scenario) that both positively and negatively affect the broader local community outside the project area which have been identified and assessed by the project, and are detailed in Section 3.2 below.

In general, the project sits within a highly developed, diversified, and mobile modern economy. Data outlined in the CCB PDD show that the local communities have a high level of economically diversity, are not highly dependent on the forestry sector, and have a low vulnerability to disruptions in the forest sector.

The CCB PDD also outlines data showing the local communities in the Darkwoods area highly value the environment, the regions natural beauty and amenities, and the related quality of life. There is evidence that the quality of the local environment attracts people and businesses to the area, drives a high level of economic diversity into the area, and may positively benefit real estate values and other positive local community features.

Summarized, the implementation of the project potentially results in a loss of economic opportunity/employment for a small number of stakeholders in the regional area due to the lower timber harvest volume in the project versus the baseline scenario (particularly in the first 15 years of the project). However, the implementation of the project also creates an offsetting amount of direct economic benefit for stakeholders and the broader local community due to property planning and management, project monitoring, project activities (such as harvesting, silviculture, and road deactivation work, etc.), increased recreational activity opportunities, and other project activities and expenditures generally benefiting the local or regional community. Further, the project results in substantial climate services benefits, ecosystem services provisioning benefits, and community amenity value impacts that benefit the local community and stakeholders.

Overall, the project creates a net benefit to local communities as summarized in Section 3.4 below.

3.2 Expected Impacts on Stakeholders

Impact #1	Reduced economic opportunity from avoided timber harvesting.
Type of Impact	Negative, direct, estimated impacts. The avoidance of the higher levels of timber harvesting in the baseline scenario reduces the employment opportunities in the logging industry. The impact is estimated using regional economic modeling of timber related employment applied against the projected reduction in timber harvesting volume from the baseline scenario.

Affected Stakeholder Group(s)	Local and regional logging and log processing industry/workers.
Resulting Change in Well-being	Reduction in timber harvest volume, particularly for the first 15 years of the project result in less direct employment and economic value related to the project area. The magnitude of potential loss is estimated in Table 7 based on regional economic study (as summarized in Table 6). Over the project lifespan, implementing the project is estimated to reduce employment generated by an average of 64 person years per year of logging related employment valued at ~$2.74 million per year[6][7]. See the project CCB PD documentation Section G2.4 for additional details on the baseline scenario impacts on communities.

Table 6. Key Harvesting Employment Variables Within the Project Zone (Lefebvre, 2008; SIBAC, 2009)

Average Harvested Volume (Kootenay Lake TSA; all ownerships)	754,953 m^3/year
Estimated Direct Harvesting, Planning, and Silviculture Employment (Kootenay TSA)	258 PY (Person Years)
Estimated Direct Primary Processing Employment (Region wide; generated by Kootenay Lake harvest volume)	494 PY (Est. 249 PY within Kootenay Lake TSA)
Total Estimated Direct Forest Industry Employment	846 PY
Employment Coefficients:	
Harvesting/Silviculture (employment generated in Kootenay Lake TSA)	0.342 PY/1000m^3
Primary Processing (employment generated in Kootenay Lake TSA)[8]	0.327 PY/1000m^3

[6] Note, however, that it is unclear whether these employment opportunities related to the Darkwoods baseline timber supply would be realized by local communities due to timber supply substitution, import of service providers from outside the project zone, and/or export of this timber supply to less proximal timber processing facilities. In other words, the direct employment benefit potential from the baseline scenario might have been significantly less than expected for the local community, with much of the benefit accruing to other regional areas. To be conservative, the project has considered these as impacts on the local communities. This is discussed in further detail in the project CCB PDD.

[7] This employment impact estimate can range from ~200 person years valued at ~8.6 million per year to 0 person years generating $0 of employment income per year depending on fluctuations in projected annual timber harvesting in the baseline scenario. XX SEE TABLE XX.

[8] Lefebvre (2008) reported an employment coefficient of 0.655 PY/1000m3 for primary processing in the region, and noted it was difficult to isolate a specific TSA coefficient due to the amount of log trading undertaken, particularly between the Kootenay Lake, Arrow, and Columbia TSA's. In order to look more specifically at employment within the project zone, we have estimated Kootenay specific employment by extrapolating data from the BC Fiber Flow reporting (BC Ministry of Forests, 2009), which used scaling data showed 50.4% of the Kootenay Lake harvest volume is processed within the Kootenay Lake TSA: Primary processing coefficient (Lefebvre, 2008) 0.655 PY/1000m3 * 50.4% (BC MOF, 2009) = 0.327 PY/1000m^3

Total employment generated in Kootenay Lake TSA from Kootenay Lake TSA harvest volume	0.669 PY/1000m^3
Average employment income from forestry and wood processing in the Kootenay Lake TSA ($2005)	$42,630/year

Comparatively, a report by SIBAC (2009) reported Statistics Canada data that the Kootenay Lake TSA had a labour force of 585 in Logging and Forestry, and 425 in Wood Products, which are relatively similar to the Person Year based calculations reported in Table 6.

Table 7. Select Economic Parameters Related to Forest Sector Activity in the Kootenay Lake TSA (Project Zone)[9][10]

	Project Average	First 15 years Average	Second 85 years Average
Baseline Harvest Volume	96,000 m^3	307,000 m^3	59,000 m^3
Harvesting Employment (PY)	33	105	20
Processing Employment (PY)	31	100	19
Total Employment	64	205	39
Employment Income ($/year)	$2,737,869	$8,755,477	$1,682,649

Impact #2	Increased economic opportunity from project activities.
Type of Impact	Positive, direct impact, predicted on an ex-ante basis, measured on an ex-post basis.
Affected Stakeholder Group(s)	Implementation of the project provides direct economic benefits to the local communities and across a wide range of stakeholders related to: 1. Project expenditures and activities 2. Project timber harvesting, silviculture, and operational activities 3. Improved recreational opportunities 4. Research opportunities

[9] See **Table 6** for the base data and assumptions used to calculate the values in this table.

[10] Note that **Table 7** reflects only direct economic impacts on workers in the project zone. There is additional indirect employment and induced employment that can be added by use of rough economic multipliers. For the purposes of this analysis, we have chosen to exclude indirect and induced employment for simplicity, due to lack of quality data, and with the assumption that by only considering direct employment/expenditures in both the project and baseline scenarios, we are making an “apples-to-apples” comparison which fairly represents the relative merits of each scenario’s impacts on local communities.

Resulting Change in Well-being	1. Project expenditures and activities NCC expenditures and projected project expenditures total an average of $550,000 per year, which is equivalent to 13 person years of employment income per year, for the duration of the project lifespan. 2. Project timber harvesting, silviculture, and operational activities NCC project timber harvesting is projected to generate on average an additional 8 person years of employment per year, valued at ~$340,000 per year for the project lifespan.

Table 8. Estimated Employment and Income Related to Project Scenario Harvesting

Project Year	NCC Harvest Volume (m^3/year)	Employment Generated (PY) (Coefficient = 0.669)	Employment Income Totals ($) (Avg. Income $42,630/year)
2008	50,763	34	
2009	40,443	27	
2010	23,819	16	
2011	11,913	8	
2012	14,152	10	
5 year average:	**28,218**	**19**	**$809,970**
100 year average	**11,943**	**8**	**$341,040**

For comparing employment impacts between the project and baseline scenarios, **Table 9** also includes an estimate of the employment related to the NCC expenditures, under the simplistic (and conservative) assumption that the wide range of contractors and consultants earn on average the same average income as the average forestry worker in the project zone from **Table 6** (likely a conservative assumption).

Table 9. Estimated Conversion of NCC Expenditures to Employment Impact

	NCC Project Expenditures ($/year)	Employment Generated (PY) (Avg. Income $42,630/year)
5 year average	$1,200,000	28
100 year average	**$550,000**	**13**

Summarized, implementation of the project scenario, at minimum, has created an average of 47 PY (19 PY from harvesting plus 28 PY from NCC project expenditures) of employment over the first 5 years of the project and projects an average of 21 PY per year over the duration of the project. The project contributed approximately $3.5 million per year of employment income and other expenditures to the local economy over the first 5 years, and is expected to average $1.36 million per year of employment income and other expenditures for the duration of the project lifetime.

Impact #3	Climate services provisioning benefits.
Type of Impact	Positive and direct impact. Actual GHG emission impacts are measured and monitored, while quantitative value is estimated from various carbon market value sources.
Affected Stakeholder Group(s)	All local community members along with national and global benefits.
Resulting Change in Well-being	The project provides an estimated net value of ~$2.4 million per year of climate services value to communities.

The ecosystems on the Darkwoods property provide significant climate benefits in the form of carbon storage and ongoing sequestration in the forest biomass. These climate benefits provide an additional quantifiable value to local (and global) communities. The avoidance of the logging development in the baseline scenario results in significant net emissions reductions over the project lifetime, as a result of reducing GHG emissions into the atmosphere and removing existing atmospheric CO_2 by way of storage in forest biomass. Current and developing carbon market and taxation schemes serve to establish a real market value on these climate services. This carbon value serves to represent the cost to communities for current and future economic and societal changes due to climate change, and to some degree represents the potential cost of climate adaptation.

By assessing the economic value of the GHG emissions related to the baseline and project scenario under various market pricing scenarios, the economic value of the Darkwoods project to communities can be assessed and estimated[11]. We have selected three representative carbon price data sources to represent a range of climate services benefit valuation[12][13]:

B.C. Carbon Tax Pricing. The B.C. Government has legally established a carbon tax that prices GHG emissions at $30.00/tCO$_2$e, which had been growing annually until a recent temporary price freeze (http://www.fin.gov.bc.ca/tbs/tp/climate/A4.htm). B.C. has also passed climate neutral government legislation that sets an carbon offset price of $25.00/tCO$_2$e for internal offset pricing via the Pacific Carbon Trust (http://www.env.gov.bc.ca/cas/legislation/index.html). For the purpose of this simple analysis, $25.00/tCO$_2$e on a real basis is representative of this provincial valuation of climate services.

Voluntary Carbon Markets. Forest Trends (Peters-Stanley, M. et al, 2013) reports that the average market price for forest carbon projects is $9.20/tCO$_2$e. NCC has successfully sold

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[11] Climate services affect the entire global atmosphere, however, for the purpose of this analysis we have assumed the benefits are apportioned to the local communities from which the benefits are realized.

[12] For consistency with the data shown in the project's CCB PD we have retained the original carbon price assumptions for this assessment. In reality, these price indicators are lower than current (2019) prices in each of the three examples. These lower prices suitably demonstrate the estimated economic value of the climate services on Darkwoods for the purposes of this SD VISta certification as well.

[13] For the simple valuation analysis undertaken here, we are assuming a constant carbon price as a real (sans inflation) snapshot into valuing carbon offsets in order to represent the value of climate services to communities. This simplistic price snapshot approach is conservative, as carbon values in virtually all market conditions are projected or legislated to rise at a rate well above inflation, and actual climate services value from Darkwoods would be significantly higher under increasing price valuation models.

VCUs from the Darkwoods project at between \$6.00-\$15.00/tCO_2e. A representative mid-range price of \$10.00/tCO_2e can used as a market value proxy of the voluntary markets current willingness to pay for climate services at the project level.

California Air Resource Board Cap and Trade Program (CA ARB). A third scenario for the valuation of climate services is the recently implemented CA ARB regulatory Cap and Trade Program. As the most recent very large regulatory carbon market, California ARB allowance prices can provide another comparative real market price on the climate services provided by Darkwoods. In 2013, California ARB Allowance prices ranged near \$14.00/tCO_2e (Point Carbon, 2013).

Table 10 shows a simple valuation[14] of the climate services provided by the Darkwoods project to communities.

Table 10. Estimated Darkwoods Climate Services Value to Communities

Net Emission Reductions by Scenario:				
	Baseline (tCO_2e/year)	Project (tCO_2e/year)	Net Change (tCO_2e/year)	Net Saleable VCU's (tCO_2e/year)
5 year Average	(335,386)	7,245	1,684,177	253,623
15 year Average	(346,645)	51,200	397,845	308,574
100 year Average	**(42,483)**	**104,126**	**146,609**	**124,847**
Carbon Pricing Scenarios:				
	BC Legistlation	Voluntary	CA ARB	
Avg. \$/tCO2e	\$25.00	\$10.00	\$14.00	
Climate Services Value Estimate Scenarios:				
	Carbon Price (\$/tCO_2e)	Baseline Net Emissions (\$/year)	Project Net Emission Reductions (\$/year)	Saleable VCU Volume (\$/year)
BC Carbon Tax Scenario	\$ 25.00	(1,062,063)	2,603,157	3,665,220
Voluntary Market Scenario	\$ 10.00	(424,825)	1,041,263	1,466,088
CA ARB Scenario	\$ 14.00	(594,755)	1,457,768	2,052,523
Average Value Impact:		**(693,881)**	**1,700,729**	**2,394,610**

The project scenario results in an estimated \$1.7 million per year of direct net carbon sequestration services to communities for the duration of the project. When combined with the climate services associated with the avoidance of carbon emissions in the baseline scenario, the project provides an estimated net value of \$2.4 million per year of climate services value to communities.

Impact #4	Ecosystem services provisioning benefits.

[14] Note that these simple calculations are intended to be representations of the order of magnitude of climate services value to communities that are suitable for the basic comparative analysis being made in this report. These values are not based upon more complex valuation techniques such as discounted cashflows, and should not be viewed as a representation of actual carbon offset or property asset values on Darkwoods or any other project.

Type of Impact	Positive and generally direct impacts. Impacts are estimated. Implementation of the project has positive impacts on: 1. Water Quality/Quantity 2. Key Species Habitat 3. Pollinator Services 4. Recreational Hunting/Fishing 5. Outdoor Recreation 6. Wetland Services/Protection These impacts can be qualitatively estimated based on regional data and studies, as shown in Table 11.
Affected Stakeholder Group(s)	All local community members along with national and global benefits.
Resulting Change in Well-being	Significantly positive across all qualitative assessments. A rough value estimate is in excess of $1.13 million/year for the project duration

As discussed in detail in the CCB PDD Section G2.4, the project provides substantial further benefits for non-climate ecosystem services provisioning. As noted, these other ecosystem services values are complex and difficult to value reliably, primarily due to the lack of established markets and valuation techniques. It is particularly difficult to measure and attribute ecosystem services provisioning at a specific location and project level. Although several studies have attempted to estimate ecosystem services value using a series of comparative techniques (i.e. Watrecon, 2010; Pagiola et al, 2004; Constanza, et al, 2006; Conservation International 2012; etc.), the ultimate empirical evidence of what markets willing to pay for particular ecosystem services values is only beginning to emerge (Forest Trends, 2008; Bennett, et al, 2013; Madsen, B. et al, 2011, etc.). Therefore, it is beyond the scope of this report to fully explore estimating quantitative ecosystem services values for the Darkwoods property; however, there is clearly a significant qualitative value impact from the baseline scenario on ecosystem services provisioning from Darkwoods that must also be considered when comparing to the traditional economic value outlined above. Table 11 outlines a qualitative assessment of non-climate ecosystem services provisioning.

Table 11. Qualitative Assessment of Non-Climate Ecosystem Services Provision Impacts (see CCB PDD Section G2.4 for discussion)

Ecosystem Service	Baseline Impacts	Project Impacts	Magnitude of Value
Water Quality/Quantity	-	++	Significant[15]
Key Species Habitat	--	++	Significant[16]

[15] In particular, the Darkwoods property contributes significantly to the valuable Kootenay Lake fishery.

[16] Examples include: the endangered Mountain Caribou are an important species with extensive regional planning efforts underway outside the Darkwoods property; Grizzly Bears are regionally important keystone species, with

Pollinator Services	-	+	Low-Moderate[17]
Recreational Hunting/Fishing	+[18]	++[19]	Low to Significant
Outdoor Recreation	-	+	Moderate
Wetland Protection[20]	-	+	Low

As seen in Table 11, the project provides significant community value from non-climate ecosystem services provisioning. Water quality is of particular importance to local communities, due to the project area providing significant watershed flows into Kootenay Lake and at least two major Bull Trout breeding streams contributing significantly into the multi-million dollar Kootenay Lake recreational fishery (Andrusak and Andrusak, 2012). The provisioning of key species habitat is also of particular importance to local communities, as demonstrated (for example) by major government programs focused on regional endangered caribou habitat planning, management, and monitoring. Although difficult to value in monetary terms specifically for Darkwoods, the collective provisioning of these multiple ecosystem services may be on par or exceeding the values calculated for climate services (as shown in **Table 10**). As a comparative example, Forest Trends (2012) estimates voluntary market water, fisheries and biodiversity market values ($85 million combined) are approximately 47% of the value of the voluntary forest carbon market ($182 million/year)[21]. Therefore, if a similar ecosystem services-to-forest carbon market value ration were applied to Darkwoods, the non-climate ecosystem services value could be valued at an estimated $1.13 million/year for the project duration.

Impact #5	Community amenity value benefits.
Type of Impact	Impact is positive, indirectly created, and estimated.

Darkwoods contributing significantly to habitat, breeding populations, and migration. Darkwoods has two key drainages that contribute significant critical breeding habitat for Bull Trout. And etc.

[17] It is likely the fully functional natural ecosystems on Darkwoods contribute native pollinator habitat, breeding populations, and migration; which supports diverse pollination services important to natural ecosystems and nearby agricultural areas (i.e. Creston valley area).

[18] Some key game species (i.e. deer) may benefit from improved browse habitat created by logging in the baseline scenario, although extensive logging may reduce other habitat attributes for such species. It is unclear whether access for hunting in the baseline scenario would have been allowed, as regional property practices are mixed. To be conservative, a positive impact rating is given.

[19] In some cases, the project would benefit key game species such as deer and elk (browse in harvested areas, winter cover, predation cover, etc.). The benefit to recreational fishery on Kootenay Lake may be significant. NCC has allowed public hunting on parts of the property (whereas the previous owner excluded hunting access).

[20] Significant wetlands are generally protected by minimum legal operating practices, and hence would have minor impacts from either scenario; however the drainages into and out of wetlands and small wetlands may be impacted by logging in the baseline scenario, leading to a negative overall impact rating.

[21] Note that when Forest Trends (2012) includes government-run ecosystem services programs, the water, fisheries and biodiversity markets exceed $45 billion, demonstrating the eventual value of ecosystem services provisioning to society.

Affected Stakeholder Group(s)	Local communities (all).
Resulting Change in Well-being	The local communities place a high value on the environment and related recreational and quality of life values. The Darkwoods project contributes to these values and improves the quality of life, livability, and desirability of living in the local area (i.e. a qualitative positive impact through improved amenity value for local communities).

Community Amenity Value Impacts

As detailed in Section G1 in the project's CCB PD, the project zone and surrounding region place very high value upon environmental values; consequentially, the conservation of large-scale properties in the project zone can materially contribute to the retention or even improvement of the amenity value of the project zone communities. Amenity values can include local factors that generally contribute to a high quality of life, livability, or desirability of living in the local communities. Studies by the USDA ((i.e. Green, G.P., 2005; Garber-Yonts, B.E., 2004, etc.), have found that the presence of forests and wild areas can contribute significantly to: community character, cohesiveness, cultural and economic diversity, resource dependence, attractiveness to business, quality of life, and civic leadership. Similar studies have concluded the logical conclusion that communities that are highly desirable can experience higher residential and commercial real estate value, better real estate liquidity, and better community economic conditions (due to factors like attracting diversified new businesses or more talented people, improved community stability/resilience, etc.). Within the project zone, communities such as Nelson can be seen to be building many of these community features over the past several years as the community has moved away from resource industry reliance (i.e. forestry).

Although it is difficult to isolate and estimate a monetary value for the contribution of the Darkwoods project to these amenity values, it can be safely concluded that the conservation of a large scale property proximal to key communities such as Nelson and Creston would contribute to improving the amenity values in the region, particularly in comparison to implementation of the baseline scenario.

3.3 Stakeholder Monitoring Plan

Community Impact Monitoring Plan

The primary objective of the Darkwoods community monitoring is to ensure that the project continues to provide the key employment, climate services, ecosystem services provisioning, and community amenity values as described in Section CM.

Community Impact Monitoring Plan:

Tracked annually and reported for each verification period.

Monitoring Activity 1 - Economic/employment Impact:

Objective: Track and report on NCC economic investment and spending related to the Darkwoods project implementation. Demonstrate that NCC continues to provide diversified economic benefit to the community and region through project related activities and monitoring.

Tasks:

1. Generate and summarize NCC accounting records for expenses related to Darkwoods.
2. Analyze comparative employment value estimates between the baseline scenario and the project expenses during the current verification period.

Monitoring Activity 2 – Climate Services Impact:

Objective: Track and report on the achieved net emission reductions generated by the Darkwoods project in the verification period.

Tasks:

1. Report on the most current ex-ante projections for net emissions from the relevant baseline and project years, and updated relevant average carbon value measures (i.e. B.C. carbon tax and/or offset pricing; CA ARB or Quebec allowance pricing, voluntary market sources, or other justifiable sources).
2. Report on the VCUs projected and issued, along with approximate sales proceeds and prices (to be provided on a confidential basis to verifiers only).
3. Calculate an average estimated climate services value for the verification period.

Monitoring Activity 3 – Other Ecosystem Services Provisioning Impact

Objective: Qualitatively (or quantitatively, if relevant data is available) re-assess the provision of other ecosystem services provisioning by the project in the verification period in comparison to an analysis of the potential baseline scenario impacts.

Tasks:

1. Report on any material project area landuse changes that impact key ecosystem services provisioning
2. Update qualitative value assessments for other ecosystem services in the baseline and project scenario years relevant to the verification period.

Monitoring Activity 4 – Community Amenity Value Impacts

Objective: Qualitatively (or quantitatively, if available) re-assess the provision of positive community amenity value from the project in the verification period in comparison to an analysis of the potential baseline scenario impacts.

Tasks:

1. Report on any material project area landuse changes that impact positive community amenity value.
2. Report on any updated community quality of life or other survey or community information that reports on the value of surrounding area environmental values (for example, references to the importance of water & air quality, recreation, scenery/aesthetics, etc.).

Key Outcome: demonstrate that, on average, the project continues to deliver net positive community benefits in comparison to the baseline scenario by updating the key quantitative and qualitative value assessments made in Section CM1, within each verification period.

3.4 Net Positive Stakeholder Well-being Impacts

Net Community Impacts of the Project

The project has a net positive community impact, based on estimates of the net value impacts of the baseline scenario compared to the project scenario. The baseline scenario provides additional direct employment opportunity (particularly in the initial 15 years); however it is not clear whether all of these employment opportunities will be realized due to timber supply substitution, import of service providers from outside the project zone, or export of the additional timber supply to facilities outside the project zone. The project scenario does provide a lower level of direct employment opportunities, however the project also provides additional employment in fields such as recreation and scientific research on the project area that have not been included here.

As would be expected, the project provides a similarly lower employment income benefit as compared to the baseline. However, the project scenario provides a significant climate services value, which is estimated to strongly outweigh the employment income in the baseline scenario. In other words, the combination of employment income, project expenditures, and climate services provides as much as 1.8 times higher combined economic benefits to the communities in the project scenario as compared to the baseline scenario. And then the project further provides clearly higher net value to communities in terms of the qualitatively assessed non-climate ecosystem services provisioning and in community amenity value[22].

Table 12. Comparative Summary of Community Benefit Estimations by Scenario (Annual Averages)

	Baseline Scenario				Project Scenario				Result
	PY	$M	# Pos.	# Neg.	PY	$	# Pos.	# Neg.	
Employment	**64**				**21**				3x Higher Employment (Baseline)
Employment Income/Other Spending		$ 2.74				$ 1.36			
Climate Services Value		$ (0.69)				$ 2.39			
Sub-Total Economic Value:		**$ 2.05**				**$ 3.75**			1.8x Higher Economic Spending (Project)
Ecosystem Services (net count)			1	-6			9	0	
Amenity Servies			0	-1			1	0	
Net Services Impact Total:				**-6 (-)**				**10 (+)**	2x Higher qualitative assessment (Project)

In total, the project provides net positive benefits to People and Prosperity through economic opportunities, climate impact value, and various ecosystem services provisioning value.

[22] The majority of the positive impacts of the project for communities are by nature shared equitably in the local communities (i.e. climate services, ecosystem services provisioning, and community amenity value). Direct economic benefits are more targeted to particular skillsets and technical expertise (similar to the baseline); however, over the life of the project these benefits are equitably available to any community member who acquires the pre-requisite capabilities. Further, the project has significantly increased the diversity of employment opportunities available as compared to the single logging employment in the baseline, which implies a more diversified opportunity for local community members of various skillsets to participate.

4 BENEFITS FOR THE PLANET

4.1 Condition of Natural Capital and Ecosystem Services at Project Start

The Darkwoods project conserves and contributes significantly to natural capital and ecosystem services (as compared to the baseline scenario). Estimated natural capital and ecosystem services quantitative and qualitative value impacts to local communities are described in Section 3. In addition to these net positive impacts to local and global communities, the key elements of the Darkwoods project to natural capital and ecosystem services is through the protection and enhancement of biodiversity and ecosystem function across the project area.

Project Zone Biodiversity Description:

Current biodiversity within the project zone is very high. For example, to date, 200 animal and 219 plant species have been confirmed on the Darkwoods property (Steeger and Machmer 2009). This includes three IUCN red-list species: Grizzly Bear (*Ursus arctos*)(threatened), Wolverine (*Gulo gulo*)(near threatened), Bull Trout (*Salvelinus confluentus*) (vulnerable). In total, 19 globally, nationally, and/or provincially significant species-at-risk are predicted to occur on the property, 15 of which have been confirmed (Steeger and Machmer 2009). Darkwoods likely provides significant habitat for at least 7 of these species-at-risk, including Grizzly Bear (Special Concern; Blue list), Mountain Caribou (*Rangifer tarandus caribou*) (Threatened; Red list), Olive-sided Flycatcher (*Contopus cooperi*) (Threatened; Blue list), Peregrine Falcon (*Falco peregrinus*) (Special Concern; Red list), Western Toad (*Bufo boreas*) (Special Concern; Yellow list), Wolverine (Special Concern; Blue list) and Bull Trout (Blue list)(NCC 2011).

Biogeoclimatic subzones include the Engelmann-Subalpine Spruce (ESSF) dry-mild (dm) and wet-cold (wc) subzones, as well as the Interior Cedar-Hemlock (ICH) dry-warm (dw) and moist-warm (mw) subzones. Ninety-seven site series (ecosystems within these subzones) have been mapped as part of the baseline inventory on the Darkwoods property. The project zone includes the Creston Valley Wildlife Management Area, which has been designated a RAMSAR site i.e., it contains wetlands are of international significance in terms of ecology, botany, zoology, limnology and hydrology.

Site conditions and natural disturbance from fire, forest pathogens, wind, and other disturbances created a diversity of forest types, many of which are very old (>100 years of age). Man-made disturbances from timber harvesting, fire, roads and historical mining, however, have disturbed the project area over the past 100 years. Threats to biodiversity (see Table 13) were identified by NCC using a method adapted from Harcombe (2007) and assessed using a simple threat rating system developed by The Nature Conservancy, World Wildlife Fund, and Foundations of Success (World Wildlife Fund 2007). The system assigns threat ratings based on an assessment of scope, severity, and irreversibility for each biodiversity target, following IUCN (2006) standards. Since the Libby Dam was built in 1975, hydrologic alterations have eliminated seasonal flooding (important for maintaining wetlands), lowered groundwater levels, and degraded wetland habitats in the region.

Table 13. Definitions of Threats Associated with Biodiversity on Darkwoods

Threat	Definition
3.2.1 Undersurface Rights Development	Habitat destruction and degradation from Mineral exploration and development
11.1.1 Global warming	Global warming may result in altered hydrological regimes, such as less snow, earlier run-off, etc.
7.1.1 Landscape Disturbance from Fire	Fire that may destroy older forests
5.3.1 Forest Harvesting	Commercial forest-harvesting activities that may result in new roads, animal disturbance, and introduction of noxious weeds
8.1.1 Invasive Species	Habitat loss and degradation from noxious weeds and other invasive alien plants and animals
1.3.1 Recreation Development	The Tye site and other infrastructure used by recreationalists that may displace animals, cause overfishing, or introduce noxious weeds
6.1.1 Recreation activities	Commercial and non-commercial recreation such as snowmobiling, fishing, skiing that may that may displace animals, cause overfishing, or introduce noxious weeds
4.1.1Roads	Existing and new logging roads may be prone to erosion and slumping, as well as providing access to sensitive areas
8.2.1 Forest health	Forest pests such as mountain pine beetle and other pathogens impact forest health
8.2.2 Caribou Predation	Predators such as wolves and cougars will kill caribou, using expanded road networks for access

High Conservation Values

The project zone includes parts of six conservation areas identified in the Canadian Rocky Mountains ecoregional assessment: Salmo River, Slocan River, Salmo/Priest/Selkirks, Kootenay River B, Kootenay River A, and Lower Columbia C (Harcombe 2007). It contains the Canadian portion of habitat for the southernmost herd of Woodland Caribou in North America (Hatter 2000), a threatened Grizzly Bear population unit (Proctor 2011), and has the highest tree diversity in British Columbia (Braumandl and Curran 1992). It includes the Creston Valley Wildlife Management Area (CVWMA), designated a RAMSAR site (wetlands that are of international significance in terms of ecology, botany, zoology, limnology and hydrology: https://www.crestonwildlife.ca/about/designations). The last remaining western breeding site for the Northern Leopard Frog (*Lithobates pipiens*) is found in the CVWMA. Large numbers – hundreds of thousands - of waterfowl pass through the valley during spring and fall migration; single day concentrations may exceed 40,000 on occasion. The area supports significant populations of grebes, herons, geese, ducks, and terns. It is the only known breeding site for Forster's Tern in B.C. The marsh supports one of the few breeding population of Black Terns in B.C., which have experienced population declines across their entire range. A continuing loss of habitat due to wetland drainage is the main reason for the decline in Black Tern populations (http://www.npwrc.usgs.gov/resource/wildlife/nddanger/species/chlinige.htm)

Federally listed species in the area include Mountain Caribou[23](Threatened), Western Screech-owl (*Megascops kennicottii macfarlanei*) (Endangered), Yellow-breasted Chat (Endangered), Badger (Endangered), Western Skink (Special Concern), Giant Helleborine (Special Concern), Short- eared Owl (Special Concern), White Sturgeon (Kootenay River population) (Endangered), Shorthead Sculpin (Threatened), and Columbia Sculpin (Special Concern)(Harcombe 2007). There are at least 20 other plant and animal species on the provincial red and blue list, but which have not yet been assessed by COSEWIC for federal listing. Of special interest is the Creston Northern Pocket Gopher (*Thomomys talpoides segregatus)*, a subspecies known only from one valley north of Creston (see Harcombe 2007).

Landscape Level Ecosystems

Darkwoods is situated in an area of high ridges and mountains interspersed with wide valleys. This area comprises part of the Interior Cedar Hemlock (ICH) biogeoclimatic zone, and encompasses an area of about 5 1⁄2 million hectares. Commonly referred to as the "interior wet belt" or "inland rainforest", the ICH contains the most productive forests of British Columbia's Interior and more tree species than any other ecological zone in the province. The lower elevation forests are characterized by Western Hemlock (*Tsuga heterophylla*), Western Red Cedar (*Thuja plicata*) and the increasingly rare, Western White Pine

[23] The term "Mountain Caribou" used herein refers to the ecotype of Woodland Caribou that occupies southeastern British Columbia. Edmonds (1991) suggested referring to these caribou as the " Mountain/Arboreal ecotype" while Thomas and Gray (2002) referred to them as the "arboreal lichen-winter feeding ecotype. Both are attempts to avoid confusion with the popular name "mountain caribou," which has been applied to caribou occupying mountains in other jurisdictions in Canada

(*Pinus monticola*). On drier sites Douglas-fir (*Pseudotsuga menziesii*), Western Larch (*Larix occindentalis*) and Ponderosa Pine (*Pinus ponderosa*) can be found.

Old forests predominate in wetter parts of the zone where fires are infrequent. Favorable conditions mean that trees grow to sizes and ages rivaling the old growth forests of coastal British Columbia. Forests in this zone contain many standing dead trees (snags) and large accumulations of fallen logs and other woody debris. These features of old forests provide valuable habitat for a wide variety of life forms, from seedlings and fungi to birds and bears. Some of the province's best Grizzly and Black bear habitat is located here.

Threatened or Rare Ecosystems

The hydro-riparian ecosystems (rivers, streams, creeks; riparian areas) in the project area are ecologically diverse and face a variety of threats. The size of contiguous segments of these ecosystems has often been restricted by past land use in the region, particularly logging, urban development, dams and hydro development, and pollutants from mining and agriculture. Dry Interior Cedar-Hemlock forests (ICHxw and ICHdw) are a rare forest type degraded by forest harvesting, range activities (cattle grazing), and fire suppression. Past logging has reduced the distribution and abundance of old-growth forests.

Critical Ecosystem Services

The Darkwoods property provides a series of important ecosystem services to the region, including hydrological services for water quantity and quality in watersheds feeding into the Kootenay Lake and the Columbia River Basin, along with other important watersheds to the west. Bull Trout are listed as a species of special concern in B.C. and Darkwoods provides critical breeding habitat for this species in the form of two key drainages (Midge Creek and Cultus Creek). Estimates suggest that more than 12,000 Bull Trout are caught annually in Kootenay Lake. Darkwoods also provides important habitat and refuge for rare and keystone species in the ecosystem, including Caribou, Grizzly Bears, Wolverine, and Western White Pine. The property likely supplies nesting habitat for the native insect pollinators that provide supporting services to agricultural operations in the Creston Valley. Climate-related benefits are accrued from the CO_2 sequestered in the forests of Darkwoods.

Traditional Use

As noted, the Darkwoods project area is a private land is within the Traditional Territory of the Ktunaxa Nation. Although the property may see occasional general use by Ktunaxa members in a manner similar to the general forest region as a whole, there are no known areas within the project area that are critical for traditional use, nor any areas receiving regular, ongoing, or special use (Choquette, 2010). In particular, there are no known areas of traditional use that are negatively affected by implementation of the project.

Exceptional Biodiversity Benefits

The Darkwoods property is a site of global significance for biodiversity conservation. With a total area of 55,200 ha, the size of the property is significant. The current and long-term biodiversity potential is very high. To date, 200 animal and 219 plant species have been confirmed on the property (Steeger and Machmer 2009). Ninety-seven site series (ecosystems) have been mapped for the property (NCC 2010) for the baseline inventory.

IUCN red-list species found on the property to date are: Grizzly Bear (*Ursus arctos*)(Threatened), Wolverine (*Gulo gulo*) (Near Threatened), Bull Trout (*Salvelinus confluentus*) (Vulnerable). Other notable species: In total, 19 Globally, Nationally, and/or Provincially significant species-at-risk are predicted to occur on the property, 15 of which have been confirmed (Steeger and Machmer 2009).

Darkwoods likely provides significant habitat for at least 7 of these species-at-risk, including Grizzly Bear (Special Concern; blue list), Mountain Caribou (*Rangifer tarandus caribou*) (Threatened; Red list), Olive-sided Flycatcher (*Contopus cooperi*) (Threatened; Blue list), Peregrine Falcon (*Falco peregrinus*) (Special Concern; Red list), Western Toad (*Bufo boreas*) (Special Concern; Yellow list), Wolverine (Special Concern; Blue list) and Bull Trout (Blue list).

The biodiversity significance of Darkwoods is partly a consequence of its geographic extent but is also related to its topographical gradient. From an elevational perspective, the property extends from warm, moist valley bottoms, to cold and dry, high elevation Engelmann spruce/Subalpine fir forests. Under the warmer climate predicted for the region, many lower elevation species are likely able to migrate upwards as their original habitat deteriorates while upslope conditions begin to ameliorate.

In addition to the magnitude of the area encompassed by Darkwoods, the property has considerable importance because of its adjacency to, and connectivity with, three provincial protected areas. These are West Arm Provincial Park (WAPP), Midge Creek Wildlife Management Area (MCWMA), and the Creston Valley Wildlife Management Area (a 7,000 ha RAMSAR site; see s. G1.8). WAPP is an area of about 25,300 ha that extends along the shore of Kootenay Lake from Nelson to Harrop, BC and up to the peaks behind these towns As the natural habitat for several endangered species, the area was protected for its biodiversity value and which also serves to protect Nelson's water source. The MCWMA is a 15,163 hectare protected area that provides critical lower elevation wildlife habitat for a variety of species-at-risk, including grizzly bear and mountain caribou. In totality, Darkwoods constitutes about 50% by area of a contiguous protected area of about 100,000 ha.

4.2 Expected Impacts on Natural Capital and Ecosystem Services

NCC has identified key biodiversity and natural capital targets for Darkwoods that are monitored and reported in annual Effectiveness Monitoring (Table 14 and Table 15). Overall property management and biodiversity targets are described in the Darkwoods Property Management Plan, which follows adaptive management principals as it is updated every five years. The primary and current key indicators of effectiveness monitoring include:

Impact #1	Dry interior cedar-hemlock forest
Type of Impact	Positive, actual, direct impacts. Old seral stage dry ecosystem forests (ICHdw) are important to key habitat and ecosystem function on Darkwoods, and hence managing to maintain or increase the proportion of the property area in these old forest conditions is important to biodiversity values.
Affected Natural Capital and/or Ecosystem Service(s)	Proportion of ICHdw that is > 140 years old.
Resulting Change in Condition	Net positive – see Table 15.

Impact #2	Grizzly Bear
Type of Impact	Positive, actual, direct impacts. Grizzly Bears are important capstone species regionally.
Affected Natural Capital and/or Ecosystem Service(s)	1. Proportion of core in Structural Stage 3 2. Proportion of core with no access/non-motorized access

Resulting Change in Condition	Net positive – see Table 15.

Impact #3	Mountain caribou
Type of Impact	Positive, actual, direct impacts.
Affected Natural Capital and/or Ecosystem Service(s)	Proportion of capable habitat lost due to anthropological means.
Resulting Change in Condition	Net positive – see Table 15.

Impact #4	Hydro-riparian ecosystems
Type of Impact	Positive, actual, direct impacts.
Affected Natural Capital and/or Ecosystem Service(s)	1. Barriers to fish flow/passage on the property 2. Proportion of watersheds with Equivalent Clearcut Areas of <20% 3. Number of bridges on the property at high risk of failure 4. Kilometers of roads rates at medium or high risk of failure
Resulting Change in Condition	Net positive – see Table 15.

Impact #5	Old growth cedar-hemlock forest (ICH types)
Type of Impact	Positive, actual, direct impacts. Old seral stage ecosystem cedar-hemlock forests are important to key habitat and ecosystem function on Darkwoods, and hence managing to maintain or increase the proportion of the property area in these old forest conditions is important to biodiversity values.
Affected Natural Capital and/or Ecosystem Service(s)	Proportion of ICH that is > 140 years old.
Resulting Change in Condition	Net positive – see Table 15.

Impact #6	Rare ecosystem and features
Type of Impact	Positive, actual, direct impact.
Affected Natural Capital and/or Ecosystem Service(s)	Proportion of rare ecosystems and features protected.
Resulting Change in Condition	Net positive – see Table 15.

Impact #7	Invasive species
Type of Impact	Risk, predicted, direct. Invasive species present a risk to natural ecosystems and hence the project maintains an active invasive species monitoring and removal program.
Affected Natural Capital and/or Ecosystem Service(s)	Number of invasive species occurrences and removals on the property. .
Resulting Change in Condition	Net positive - see Figure 3.

No known invasive species have been introduced to Darkwoods as a result of project activities. Invasive species occurrence data are available from the Invasive Alien Plant Program (IAPP) (https://catalogue.data.gov.bc.ca/dataset/invasive-alien-plant-site), run by the BC Government's Ministry of Forests, Lands and Natural Resource Operations (FLNRO). A total of 8 species have been recorded during the previous monitoring periods on the project area, and the number of species has not been increasing during the project. NCC maintains an active invasive species removal program, including . monitoring for the the presence of invasive species on a periodic basis across the property and removal of known invasive species on a periodic or annual basis by chemical, biological, or mechanical means. Figure 3 shows the occurrence of invasive species removal activities during a previous monitoring period.

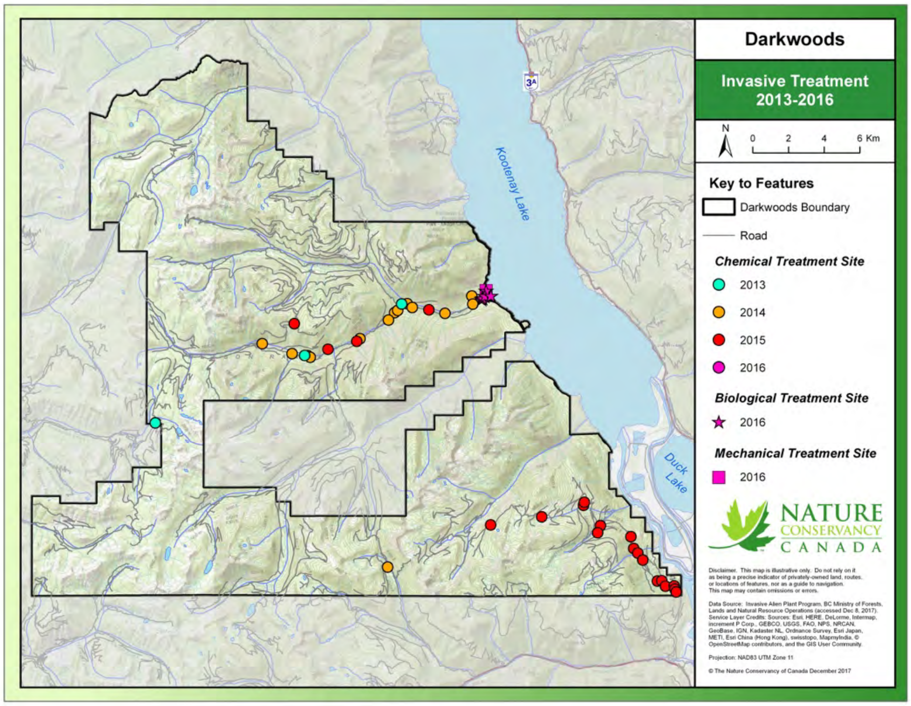

Figure 3. Darkwoods Invasive Species Removal Program 2013-16

Note also the impacts on Natural Capital and Ecosystem Services discussed in Section 3 above, in particular the estimation of a quantitative and/or qualitative assessment of climate and ecosystem services provisions outlined in Impact #3 and Impact #4 (Section 3 Impact Tables).

4.3 Natural Capital and Ecosystem Services Monitoring Plan

Biodiversity monitoring activities and processes on the Darkwoods property are designed to provide sufficient information to reduce threats and meet biodiversity goals (NCC 2011), and to be adaptive to new information, objectives, and targets over time. Detailed discussion of the biodiversity threats, targets, and related management objectives and activities are described in the Darkwoods Property Management Plan (PMP) (NCC 2011, NCC 2016). The Darkwoods PMP is reviewed and updated every five years and are designed to enable adaptive conservation management (and hence, biodiversity targets and related monitoring are expected to be modified and updated on a regular basis throughout the project).

The current targets and activities for biodiversity monitoring are termed “Effectiveness Monitoring”. The current project Effectiveness Monitoring is summarized in Table 14 and the most recent reported results shown in Table 15.

Table 14. Summary of Effectiveness Monitoring - Darkwoods (NCC, 2013)

Summary of Effectiveness Monitoring (NCC, 2013)

					Indicator Rating Thresholds (Viability / Threat Magnitude)												
MOS-E Category	Key Attribute	Indicator	Indicator Category	Monitoring Scale	Poor/ Very High	Fair/ High	Good/ Medium	Very Good/ Low	Initial Rating (Year)	Current Rating (Year)	Estimated Trend (+, -, =, ?)	Desired Future Rating	Level of Confidence	Monitoring Methodology/ Action Step	Monitoring Frequency	MOS-E Description	Aligned PMP Action
Targets																	
Mountain Caribou	Intact suitable Mountain Caribou habitat	Proportion of suitable Mountain Caribou habitat lost to disturbance	Condition	property	>10%	5% - 10%	<5%	0				Very Good (0)	Fair	review of harvesting and natural disturbance	Every 3 years	Less than 5% of suitable Mountain Caribou habitat was not disturbed (natural or anthropogenic disturbance)	2.1.1, 1.2.1, 2.1.2
Hydro-riparian ecosystems	Water Quality	Number of high risk bridges	Condition	property	>5	3 - 5	1 - 2	0				Very Good (None)	Fair	field-based assessment	Every 3 years	All high risk bridges were removed and high risk roads were deactivated	
Hydro-riparian ecosystems	Water Quality	Kilometres of high risk roads	Condition	Property	>50km	25km - 50km	1km-25km	0				Very Good (0)	Fair	field-based assesment, review of road deactivation activities	Every 3 years	All high risk bridges were removed and high risk roads were deactivated	
Hydro-riparian ecosystems	Water Quality	Proportion of watersheds with equivalent clear-cut areas of less than 20%	Condition	property	<50%	50% - 75%	75% - 95%	>95%				Very Good (>75%)	Fair	review of harvesting	Every 3 years	Greater than 75% of watersheds on the property had an equivalent clear-cut area of less than 20%.	2.1.1, 1.2.1
Dry ICH forest	Age Structure	Proportion of dry ICH (ICHdw) greater than 140 years	Condition	property	<10%	10% - 15%	15% - 21%	>21%				Fair (10%)	Fair	review of harvesting and natural disturbances	Every 3 years	10% of ICHdw was greater than 140 years old.	1.2.1, 2.1.1, 1.2.1, 5.3.1
Grizzly bear	Suitable Habitat	Proportion of core habitat in structural stage 3	Condition	property	<10%	10% - 15%	15% - 20%	>20%				Very Good (>20%)	Fair	review of harvesting and natural disturbance	Every 3 years	Greater than 20% of core Grizzly Bear habitat was in structural stage 3 (herb-shrub).	1.2.1
Grizzly Bear	Protection from human-related harassment.	Proportion of core habitat zoned for non-motorized and/or no access	Landscape Context	property	0 - 35%	25% - 50%	50% - 75%	>75%	Fair (2009)	Good 67.0000 (2013) 50% (2010)	=	Good (67% or greater)	Good	access management map	Every 3 years	Greater than 67% of the Darkwoods property has been zoned for non-motorized/no public access	2.1.1, 2.1.2, 2.1.6
Old-growth Cedar-Hemlock forest (ICHmw)	Age Structure	Proportion of ICHmw forest that is greater than 140 years old	Condition	property	<10%	10% - 15%	15% - 21%	>21%				Good (>15%)	Fair	review of harvesting and natural disturbances	Every 3 years	Greater than 15% of ICHmw forests were greater than 140 years old	1.2.1, 1.2.1, 2.1.1, 2.1.2
Threats																	
Other																	

Table 15. Summarized Annual Biodiversity Effectiveness Monitoring Results

Target	2013	2014	2015	2016	Methods	Source
Dry interior cedar-hemlock forest						
Proportion of old forest in ICHdw	0.09	0.09	0.09	0.09	Total forested ICHdw1 = 4328.5 ha, Total ICHdw1 > 140 yrs = 397 ha OR 9% **Forested = Spc1 not blank, remove age < 3 (NSR). No harvest within this defined area in this period.	Used Ecora VRI, Deb MacKillop BGC, GIS harvest data.
Grizzly bear						
Proportion of core in Structural Stage 3 (SS3)	0.24	0.24	0.24	0.24	strstg3 = 6244/26211 = 24%or Core is SS3	Used Ecora VRI Struct Stg
Proportion of core no	0.87	0.87	0.87	0.87	Of the 26,211 ha of core mapped habitat, 22830 ha is zoned No Access or 87% of Core is No Access	Darkwoods boundary, Primary

access/non-motorized						Zone expanded
Mountain caribou[24]						
Proportion of capable habitat lost to disturbance	0.00	0.00	0.00	0.00	No logging in caribou habitat areas in the monitoring period	GIS annual harvest data
Hydro-riparian ecosystems						
Barriers to fish flow/passag e	0.00	0.00	0.00	0.00	No new activity in period	Staff reporting of activities
Proportion of watersheds with Equivalent Clearcut Areas of <20%	1.00	1.00	1.00	1.00	See annual EM report worksheets	Used carbon VRI resultant
Number of bridges on the property at high risk of failure	0.00	0.00	0.00	0.00	6 Bridges removed since 2009, 1 installed	Staff reporting of activities
Kilometres of roads rated at medium- or high risk of failure	0.00	0.00	0.00	0.00	Up to Fall 2015 - 218 km deactivated out of 603 total km of road	Staff reporting of activities
Old-growth cedar-hemlock forest						

[24] Also see: Footnote **Error! Bookmark not defined.**.

Proportion old forest in ICH	0.28	0.28	0.28	0.28	Total forested ICHmw4 = 5638.3 ha, Total ICHmw4 > 140 yrs = 1601 ha OR 28% **Forested = Spc1 not blank, remove age < 3 (NSR). No harvest or fires within this defined area in period.	Used Ecora VRI, Deb MacKillop BGC, GIS harvest data.
Rare ecosystems and features						
Proportion of sites protected	1.00	1.00	1.00	1.00	No new activity in period	Staff reporting of activities

In addition, as part of the CCB PD, NCC monitors Invasive Species, as described in Section 4.3 and displayed in Figure 3.

4.4 Net Positive Natural Capital and Ecosystem Services Impacts

In addition to the net positive impacts from ecosystem services noted in Section 3, the project will have net positive impacts on biodiversity as well as significant GHG emission reductions. The results of biodiversity effectiveness monitoring in the project periods to date have demonstrated the project has positive direct impacts on all of the key indicators described in Section 4, and shown in Table 15.

Additionally, the project has validated and verified GHG emission reductions and forest biomass protection and growth as demonstrated in the VCS PD and Monitoring Reports verified to date, and summarized below. These significant climate related benefits will continue throughout the project lifepan, as demonstrated in the ex-ante projections of ecosytem carbon stocks in the project versus baseline scenario shown in Figure 4 and projected ex-ante annual net downs and net VCU generation as shown in Table 16.

Summary of GHG Emission Reductions:

The Darkwoods project area sequesters significant carbon stocks in the forest biomass located across the property. The avoidance of baseline scenario logging results in the annual retention of carbon stocks by avoiding the removal of tree biomass and subsequent decay of the resulting dead biomass and wood product waste. The site also continues to sequester additional carbon in the growth of the retained and regenerating forests across the site. These carbon stocks and their fluxes overtime have undergone detailed measurement and modeling by carbon pool, using the Verified Carbon Standard (VCS) Methodology VM0012, as documented in the Validated Darkwoods Forest Carbon Project Design Document (PDD) and as verified in subsequent VCS Monitoring Reports.

The project area starting and modeled ex-ante projected carbon (in tonnes of carbon) within key ecosystem carbon pools are shown in Table 16. Year 0 represents the starting carbon stocks for the project area. These figures are derived from detailed forest inventory data, project permanent sample plots, and robust ecosystem modeling as fully described in the validated and verified VCS PDD.

Additional detail related to the measurement, calculation and modeling of carbon stocks can be found in the Darkwoods VCS Project Design Document, Version 1.8, including all project design details, carbon stock calculations, risk assessments, monitoring plans, etc. Key reference information related to carbon stock calculations include:

- Section 2.3 for GHG pools included
- Section 4 for Baseline and Project Scenario carbon accounting calculations
- Figure 11 summarizes the starting ecosystem carbon stock and initial PDD ex-ante projections for the change in ecosystem carbon in both scenarios
- Table 24 and 25 (Appendix 3) summarizes the ecosystem carbon stocks by pool over time in both scenarios, respectively
- Appendix 5 for supporting data files, including the accompanying MS Excel file "Darkwoods Carbon Model v.8.7a", which shows the calculation of the total change in emissions and ex-ante projection of Verified Carbon Units (VCUs)

In addition, updated inventory and carbon stock changes since the project start date can be found in the verified VCS and VCS/CCB Monitoring Reports for the periods 2008-2010, 2011-2012, and 2013-2016. The most recent verified GHG emission reduction results are shown in Table 17 and Table 18.

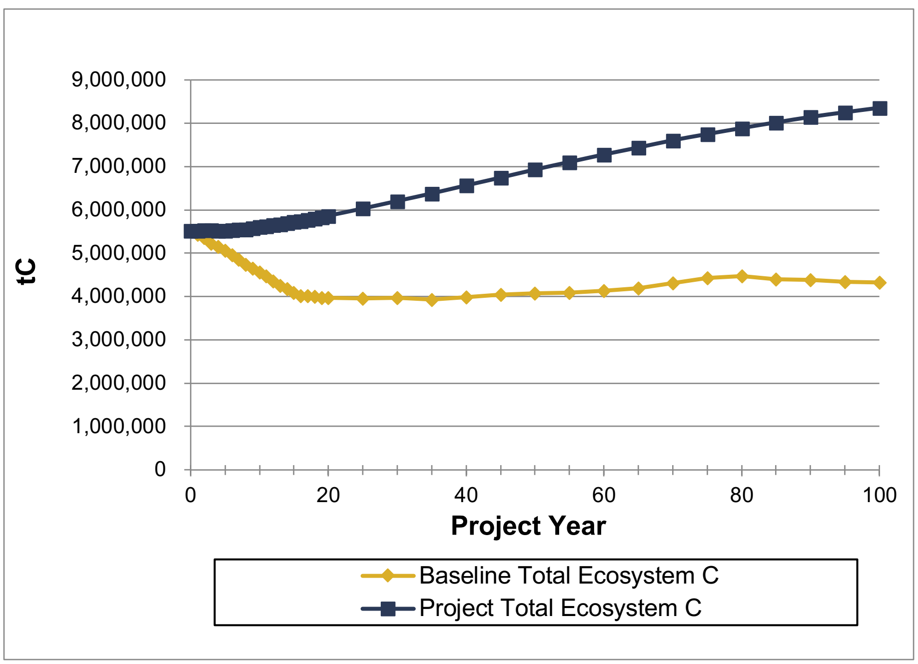

Figure 4. Ex-Ante Project Net Ecosystem Carbon Storage (Biomass + Deadwood + Belowground dead biomass) by Scenario (year 1 = 2008).

Table 16. Original Ex-Ante Net Emission Reduction Calculations (from: Darkwoods VCS PDD, Table 18)[25]

Year	Annual Net Emissions Reductions (tCO_2e)	Leakage Risk Discount (tCO_2e)	Uncertainty Risk Discount (tCO_2e)	Non-Perm Buffer Contribution (tCO_2e)	Cumulative Non-Perm Buffer Account (tCO_2e)	Non-Perm. Buffer Release (tCO_2e)	Annual Saleable VCU's (tCO_2e)
Rate		Varies	Est. 5%	10%		15.0%	
1	**317,823**	(36,587)	(14,062)	(31,463)	31,463		**235,711**
2	**331,727**	(38,123)	(14,680)	(32,840)	64,303		**246,084**
3	**439,461**	(39,709)	(19,988)	(43,600)	107,903		**336,164**
4	**244,640**	(43,387)	(10,063)	(24,086)	131,989		**167,105**
5	**350,526**	(42,384)	(15,407)	(34,683)	141,671	25,001	**283,053**
6	**436,976**	(42,312)	(19,733)	(43,329)	184,999		**331,602**
7	**428,733**	(42,528)	(19,310)	(42,502)	227,502		**324,392**
8	**439,273**	(42,316)	(19,848)	(43,558)	271,060		**333,551**
9	**401,630**	(43,653)	(17,899)	(39,782)	310,842		**300,296**
10	**435,754**	(47,233)	(19,426)	(43,163)	300,905	53,101	**379,032**
11	**403,159**	(42,913)	(18,012)	(39,942)	340,846		**302,292**
12	**452,612**	(46,849)	(20,288)	(44,853)	385,699		**340,622**
13	**413,921**	(43,106)	(18,541)	(41,016)	426,715		**311,259**
14	**430,528**	(42,425)	(19,405)	(42,683)	469,398		**326,015**
15	**440,911**	(42,816)	(19,905)	(43,718)	436,148	76,967	**411,439**
16	**290,837**	(23,032)	(13,390)	(28,883)	465,031		**225,532**
17	**131,601**	0	(6,580)	(13,183)	478,214		**111,838**
18	**126,403**	0	(6,320)	(12,654)	490,869		**107,429**
19	**243,925**	(13,649)	(11,514)	(24,273)	515,142		**194,488**
20	**134,467**	0	(6,723)	(13,459)	449,311	79,290	**193,575**
25	**127,468**	(1,066)	(6,320)	(12,647)	435,666	15,376	**122,810**
30	**116,801**	(2,736)	(5,703)	(11,511)	419,237	14,797	**111,648**
35	**158,659**	(11,713)	(7,347)	(15,303)	421,388	14,873	**139,169**
40	**97,842**	(2,384)	(4,773)	(9,629)	399,101	14,086	**95,143**
45	**81,649**	(2,758)	(3,945)	(7,987)	373,181	13,171	**80,131**
50	**117,388**	(9,127)	(5,413)	(11,291)	365,191	12,889	**104,446**
55	**119,179**	(10,841)	(5,417)	(11,395)	358,843	12,665	**104,191**
60	**90,160**	(9,184)	(4,049)	(8,561)	341,401	12,049	**80,415**
65	**74,324**	(2,308)	(3,601)	(7,280)	321,131	11,334	**72,468**
70	**41,217**	(898)	(2,016)	(4,022)	290,053	10,237	**44,519**
75	**10,230**	(1,629)	(430)	(893)	250,340	8,836	**16,114**
80	**75,355**	(15,964)	(2,970)	(6,766)	241,545	8,525	**58,180**
85	**139,373**	(16,859)	(6,126)	(13,151)	261,206	9,219	**112,456**
90	**107,151**	(8,738)	(4,921)	(10,283)	265,729	9,379	**92,588**
95	**113,304**	(10,717)	(5,129)	(10,799)	271,766	9,592	**96,250**
100	**83,095**	(6,523)	(3,829)	(7,972)	264,882	9,349	**74,120**
Total	**14,660,880**	(1,240,247)	**(671,032)**		**(264,882)**		**12,484,719**

[25] Note that these are the original ex-ante offset projection results for the project planning (PDD stage). Subsequent VCS Monitoring Reports have resulted in modifications during ex-post calculations for VCU issuance, as expected.

Table 17. Verified summary of emission removals and leakage emissions for the baseline and project scenarios for the 2013-2016 verification period.

Year	Baseline emissions or removals (tCO_2e)	Project emissions or removals (tCO_2e)	Gross GHG emission reductions or removals (tCO_2e)
2013	(241,227)	130,376	371,602
2014	(285,826)	119,893	405,719
2015	(316,975)	50,274	367,248
2016	(257,335)	157,455	414,790
Total	**(1,101,363)**	**457,997**	**1,559,360**

Table 18. Verified total saleable VCUs and terms used in their calculation for 2013-2016

Year	**Ann. Gross Emissions Reductions (tCO2e)**	**Leakage Risk Discount (tCO2e)**	**Uncertainty Risk Discount (tCO2e)**	**Annual Buffer Set-Aside (tCO2e)**	**Cumulative Buffer Account (tCO2e)**	**Release From Buffer (tCO2e)**	**Saleable VCU (tCO2e)**
2013	371,603	(44,150)	(4,912)	(35,747)	227,288		286,793
2014	405,719	(44,099)	(5,424)	(39,160)	266,448		317,035
2015	367,248	(42,433)	(4,872)	(35,367)	301,814		284,577
2016	414,790	(44,240)	(5,558)	(40,063)	290,596	51,282	376,211
Total	**1,559,360**	**(174,922)**	**(20,767)**	**(150,337)**	**290,596**	**51,282**	**1,264,617**

Overall, the project achieves net positive impacts for the Planet through biodiversity and climate related elements, in addition to the ecosystem services provisioning impacts identified in Section 3.